MRPW: Regression, poststratification, and small-area estimation with sampling weights*

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Abstract

A well-known rule in practical survey research is to include weights when estimating a population average but not to use weights when fitting a regression model—as long as the regression predictors x include all the information that went into the sampling weights w. But what if you don't know where the weights came from? We propose a quasi-Bayesian approach, which we call MRPW (multilevel regression and poststratification with weights) that estimates a joint regression of the outcome and the sampling weight, followed by poststratifcation on x and w, thus using design information within a model-based context to obtain inferences for small-area estimates, regressions, and other population quantities of interest.

1. Background

1.1. Survey weighting

One of the central challenges of statistics is generalizing from sample to population. The natural first step here is to adjust for known, expected, or assumed discrepancies between sample and population[1]. But even this basic level of correction can be challenging, especially when sample and population diverge in many dimensions (for example, age, sex, education, ethnicity, geography, and political affiliation in social surveys).

Weighting is a way to summarize an adjustment: each item in the sample gets a nonnegative weight which is intended to be proportional to its representation in the population. Population estimates can then be obtained as weighted averages of the sample. Four difficulties arise with classical survey weighting: construction of weights, uncertainty estimates, small-area estimation, and regression modeling.

Construction of weights is difficult because real-world surveys will require adjustment for many factors, and simple approaches based on poststratification or estimated probabilities of sampling often result in highly noisy weights. Noisier weights lead to losses in the efficiency of weighted estimates: the more variability that exists in the weights, the less efficient the weighted survey estimates become (Korn and Graubard, 1999). This in turn motivates more complicated approaches based on smoothing or modeling the weights, which can be done but at the cost of many choices in modeling and estimation (Little, 1991; Gelman and Little, 1998; Elliott and Little, 2000; Little and Vartivarian, 2003; Chen et al., 2006; Gelman, 2007; Chen et al., 2012, 2017; Xie et al., 2020; Si et al., 2020; Ben-Michael et al., 2024).

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[1]An example of a *known* discrepancy between sample and population would be a sample of 60 women and 40 men that is intended to represent a population that is 52% women and 48% men. An example of an *expected* discrepancy would be clusters sampled with probability proportional to a known measure of size. These discrepancies become *assumed* if the population proportions and sampling probabilities are approximate and not known.

Standard errors or other uncertainty measures with weighted averages are challenging because a set of weights is sufficient to define a weighted average but does not specify a full probability model; additional assumptions must be added beyond those implied by the weights (Lumley, 2004; Solon et al., 2015).

Small-area estimation using weights is difficult because a small area may have so few observations that no purely local estimate, weighted or otherwise, would be reasonable (Fay and Herriott, 1979; Rao, 2003). Consider, for example, a national political survey that contains five responses from Wyoming, all of whom support the Republican candidate for president. Any weighted average would result in an obviously wrong estimate of 100% Republican support in the state. Weighting is defeated by data granularity, and modeling is required.

Regression modeling with weights can work in simple settings, replacing least squares or maximum likelihood with weighted versions of these methods. Procedures exist to test whether survey weights are needed for the estimation of a given regression model (Bollen et al., 2016), and methods have been developed to minimize the impact of noisy weights on estimated regression model parameters (Pfeffermann, 2011). But the use of weights to fit models becomes more difficult when moving to more advanced multilevel, Bayesian, or regularized methods that are needed to answer complex questions in the presence of data granularity (DuMouchel and Duncan, 1983; Pfeffermann et al., 1998; Rabe-Hesketh and Skrondal, 2006; Lumley and Scott, 2017).

This is not to say that weighting-based methods are useless. Much work has gone into population inference using survey weights, and weights can be used to give robustness to generalizations in causal inference as well as sampling (Bang and Robins, 2025). Our point here is that there are no generally applicable or easy solutions to the problem of adjusting for discrepancies between sample and population, and so there are theoretical, methodological, and applied reasons for desiring a generally-applicable and unified approach to regression modeling and small-area estimation using survey weights. The approach presented here follows ideas of Särndal (1978), Kalton (1983), Pfefferman (1993), Little (2015), and others that incorporate design information into model-based inference.

1.2. Multilevel regression and poststratification

Multilevel regression and poststratification (MRP) or, more generally, regularized regression and poststratification, is an approach to survey analysis that combines modeling of the data with adjustment for nonrepresentativeness of the sample. In the basic MRP setup, an outcome y and background variables x are observed in the sample, and the distribution of x is known in the population. If the variables in x are discrete, then their interactions define poststratification cells. If the observed data are independently sampled with probabilities of selection that do not vary within poststratification cells, population inference can be performed by fitting a regression model of y on x and then averaging over the cells in proportion to their known population counts (Holt and Smith, 1979; Little, 1993).

So far, this is simply regression and poststratification. The multilevel part comes in because, given the implicit assumption of constant probability of inclusion within cells, there is a desire to poststratify on as many factors as possible, and a regression model with a large number of predictors and interactions cannot be estimated stably using least squares. Multilevel modeling is a good way to fit a regression with many predictors such as arise when modeling survey responses given demographic and geographic factors (Gelman and Little, 1997). Other approaches are possible, hence we have also used the more general term, "regularized regression and poststratification" or RRP (Gelman, 2018; Bisbee, 2019; Broniecki et al., 2021; Gopelrud, 2024). A key attribute of MRP

(or RRP) is that it allows predictions for y given values of x that are not observed in the sample, or which have such small counts in the sample that it would be impossible to make predictions for them from local data alone.

There is a growing literature on MRP and its generalizations. Challenges include obtaining good group-level predictors for multilevel regressions (so that, for example, inferences for small states in a national survey are partially pooled toward reasonable state-level estimates rather than to a national baseline); adjusting for non-census variables, in which case the population counts of the poststratification cells themselves must be estimated from the data (Su and Gelman, 2023; Li and Si, 2024); analyzing cluster samples when the cluster sizes in the population are unknown (Graubard and Korn, 2002; Stanek and Singer, 2004); and, with particular relevance to the present research, modeling unequal sampling probabilities within poststratification cells. One quick way to incorporate survey weights is to replace the observed mean response within each cell by its weighted mean and use an adjusted within-cell variance estimate (Potthoff et al., 1992; Ghitza and Gelman, 2013; Chen et al., 2014), but this approach fails when data are sparse and many cells have only a single respondent, in which case important variation in the weights can be missed.

1.3. Analyzing surveys collected by others

Textbooks on survey sampling focus on the scenario in which the data are analyzed by the same team that conducted the survey. There is some literature on the construction of sampling weights, but not much on the analysis of surveys collected by others, even though this type of *secondary analysis* is a common mode of social science research (Kish, 1992; Korn and Graubard, 1999; West et al., 2016; Heeringa et al., 2017; Haziza and Beaumont, 2017; Lohr, 2022). Publicly-available surveys typically come with weights but often do not fully explain how the weighting scheme was chosen or exactly how the weights are computed, hence it can be difficult or impossible to reproduce the procedure starting from the data (Voss et al., 1995). Unfortunately, the documentation provided by these surveys for data users, talking about the weights and other design features and how they should be used, varies tremendously in detail and usefulness (Kolenikov et al., 2020).

When conducting analyses of data collected by others, researchers are often advised to use the weights when estimating population averages but not when fitting regression models, as long as all the variables that went into the weighting are included as predictors; see, for example, Winship and Radbill (1994). This advice is useful where it can be followed, but it does not resolve the question of what to do when fitting a regression whose predictors do not include all the variables that went into the weights. In addition, it is awkward to consider averaging and regression as different problems, given that averaging is a special case of regression. For example, when estimating the average within a subgroup (for example, average responses for women or men), we might simply use the weighted average from the relevant group in the sample, but if the subgroup is small enough (for example, individual states or geographic/demographic categories in a national survey), we would want to perform small-area estimation using regression.

The literature on design-based secondary analysis of survey data is clear on the point that using correctly-specified inverse-probability weights will produce consistent and asymptotically unbiased estimates of regression parameters with respect to the sample design, even if that regression model model has been poorly specified (Pfeffermann, 1993; Korn and Graubard, 1999; Heeringa et al., 2017). But such weighted estimates can be noisy; in addition, if the weights provide little or no predictive power beyond what is in the regression predictors, than weighting can simply add unnecessary noise.

We would like to get the best of both worlds, using weighting adjustments just to the extent

that the weights add relevant information, with a model-based procedure that is coherent and does not treat averaging and regression estimation as different problems.

1.4. MRPW: Incorporating sampling weights into model-based inference

The present paper shares a general approach to analyzing surveys with weights, under the scenario that the weights have already been constructed before the analyst sees the data, as in West and McCabe (2012). This is similar to the idea of multiple imputation for public surveys, in which the organization in charge of the survey uses sophisticated methods to construct imputations, and then users can analyze the imputed datasets, taking the imputations as given (Meng, 1994; Rubin, 1996). Dividing the problem in two parts—first the construction of the weights, then the analysis of the weighted dataset—entails an inevitable loss of statistical efficiency (except in some special cases), but, as with imputation, offers practical gains of division of labor and facilitates comparability of analyses by different users of the same survey.

The approach we derive and demonstrate in this paper is a generalization of MRP, in which we expand the set of predictors x to include the sampling weights w, thus performing a multilevel regression of $y|x, w$ and constructing a poststratification table for (x, w) by estimating the conditional distribution of $w|x$ and combining it with the known population distribution of x. What makes this practical is that only one new variable is added to the multilevel model and only one new set of conditional distributions needs to be modeled to complete the poststratification. This last step needs to account for the unequal sampling probabilities, and we do this by fitting a joint probability model that includes the inverse-probability sampling weights. We do this with quasi-Bayesian inference, using a weighted bootstrap of residuals to allow for a nonparametric distribution of weights within poststratification cells. We call this approach MRPW (multilevel regression and poststratification with weights), and we demonstrate it with simulated and real-data examples.

2. A quasi-Bayesian approach to regression with survey weights

2.1. Model

Suppose we have a vector of K background variables x that are observed in the sample and whose distribution is known in the population, and a weight variable $w > 0$ and scalar outcome y that are known only in the sample. Assume the data have been sampled independently from the population with probabilities inversely proportional to the weights.[2] The poststratification cells $j = 1, \dots, J$ correspond to the possible values of x in the population; we label these as X_j, with N_j being the size of cell j in the population. We assume that the N_j's are known with certainty. In practice, these quantities may be estimated based on other large probability surveys, and this uncertainty should be addressed as part of this procedure (Dever and Valliant, 2010; Li and Si, 2024).

Our goal is to perform Bayesian inference for the population values of y, given the known background variables x. Inference for $y|x$ can then be combined to get inference for the entire population or for subgroups of interest; this is the poststratification step. For example, if we are poststratifying a national poll into 4 ethnic categories, 6 age categories, 2 sex categories, 5 education categories, and 50 states, then the number of cells is $J = 4 \cdot 6 \cdot 2 \cdot 5 \cdot 50 = 12{,}000$, and the population mean value of the outcome for white people in Alabama, for example, is the weighted average of $\mathrm{E}(y|X_j)$ over the 60 cells corresponding to that group.

[2] We use the term "sampling" here to include all factors relating to inclusion in the sample, including nonresponse; see Rubin (1976) and Brick (2013).

If there were no survey weights and we could assume equal-probability sampling, we would simply regress y on x in the sample and then use the fitted models to make predictions (with uncertainty) for the rest of the population. The challenge is that the data are sampled with unequal probabilities. We use the notation p and p_{sample} for the distributions of the population and sample, respectively; that is, we are considering the items in the population to be drawn at random from an infinite superpopulation with distribution $p(y, x, w)$, so that the sample can be considered a draw from the distribution,

$$p_{\text{sample}}(y, x, w) \propto p(y, x, w)/w. \tag{1}$$

We handle the problem of unequal sampling probabilities by modeling the joint distribution of the outcome and the weights, following Skinner (1994), Beaumont (2008), Si et al. (2015), and Léon-Novelo and Savitsky (2019):

$$\text{Model for the outcome:} \quad p(y|x, w, \theta) = p_{\text{sample}}(y|x, w, \theta) \tag{2}$$

$$\text{Model for the weights:} \quad p(w|x, \phi) \propto w\, p_{\text{sample}}(w|x, \phi), \tag{3}$$

where θ and ϕ represent the parameters in the outcome and weight models. Both models (2) and (3) are conditional on x, and we are assuming that x is known in the population. The advantage of the above formulation is that it makes clear how both models can be estimated from the sample data. Because we are doing this work using simulation, it can be thought of as a design-consistent and model-based approach to generating a synthetic population as proposed by Dong et al. (2014).

In effect, we are poststratifying on (x, w), which requires estimation of $p(w|x)$ so that we can construct the joint distribution of x and w in the population.

Three key aspects of this approach are:

- The outcome model (2), which, following the principles of MRP, can include many predictors x, the functions of w, and their interactions;
- The adjustment for the sampling weights in the transition from sample to population distributions in (3), which captures the adjustment information in the weights;
- The adjustment for w is performed using a model, rather than simply reweighting individual data points. Using a model allows the method to work with sparse data, using MRP, and the observed data are used to estimate a complete population distribution.

2.2. Quantities of interest

We consider several different inferential goals.

The *overall population mean* can be written as $\overline{Y} = \sum_{j=1}^{J} N_j \overline{Y}_j / \sum_{j=1}^{J} N_j$, where $\overline{Y}_j = \text{E}(y|X_j)$ is the population mean within poststratification cell j. We are assuming an infinite or essentially infinite population so that there is no need for a finite-population correction.

The population means will never be known, so for Bayesian inference we can write, $\overline{Y} = \sum_{j=1}^{J} N_j \text{E}(y|X_j, \theta, \phi) / \sum_{j=1}^{J} N_j$, so that uncertainty in the model parameters θ, ϕ induces uncertainty in $\overline{Y}$. Posterior simulations of (θ, ϕ) propagate to posterior simulations of the vector of $\overline{Y}_j$'s, which in turn propagate to posterior simulations of the population average, $\overline{Y}$.

This approach also applies to *individual poststratification cells* or *groupings of such cells*. For example, in political surveys there can be interest in estimating the average within a state, which uses the same poststratification formula as above, but just including a subset of cells.

Another goal of interest is the *population regression*, that is, the result that would be obtained by a regression of y on x fit to the entire population. This is a more general concept than trying to estimate a "true regression coefficient" because it does not rely on the assumption that the fitted regression is a correct generative model for the data.

We can estimate a population regression within our framework in two steps. First, we estimate the population average, $\overline{Y}_j = \mathrm{E}(y|X_j)$, within each poststratification cell j. Second, we fit a cell-level regression using these averages. For a linear model, we can simply fit weighted least squares to these J "data points," with point j getting a weight of N_j corresponding to its size in the population. For logistic regression, we can maximize a weighted pseudo-likelihood with $2J$ data points, with cell j being assigned an observation of 1 with weight $N_j\overline{Y}_j$ and an observation of 0 with weight $N_j(1-\overline{Y}_j)$. In either case, we would propagate uncertainty as described above, performing these analyses for each posterior simulation of (θ, ϕ), thus obtaining posterior simulations of the desired population regression.

The problem becomes more difficult if there is a desire to go beyond simple averages and proportions within cells. For example, in the study of political polarization there is interest in estimating the *variation* within U.S. congressional districts: when considering swing districts, there is a big difference between a population of political moderates and a population divided evenly between strong partisans of the left and right. The only general way to estimate this variation is to simulate the entire distribution of responses, not just the mean, within each poststratification cell.

2.3. Inference

Before getting into details of Bayesian inference, uncertainty, and computation, let us consider how to fit (2) and (3) using point estimation. The first step is to regress y on x and w, yielding $p(y|x, w, \theta)$. The second step is to regress w on x, again using the observed data, thus yielding $p_{\text{sample}}(w|x, \phi)$. Here we are simply taking θ and ϕ as their point estimates. Next we convert from sample to population distribution,

$$p(w|x, \phi) = \frac{w\, p_{\text{sample}}(w|x, \phi)}{\int w\, p_{\text{sample}}(w|x, \phi) dw}. \tag{4}$$

This latter expression needs to be evaluated for each value of x in the population (that is, for all the poststratification cells), hence the integral in (4) must either be determined analytically or though some fast approximation. Conceptually, though, the problem is now solved: for each poststratification cell j, we determine $p(w|X_j, \phi)$ from (4) and then average over this distribution to get the predictive distribution of y in cell j:

$$p(y|X_j, \theta, \phi) = \int p(y|X_j, w, \theta) p(w|X_j, \phi) dw. \tag{5}$$

This integral can be derived analytically or else approximated. In any case, we now have estimated the population predictive distribution within each cell and can then poststratify by averaging over the assumed-known cell counts N_j in the population.

Bayesian inference is performed in the same way, with the only difference being that inference about θ and ϕ is propagated through (4) and (5). Here is a generic computational implementation:

1. Define a prior distribution, $p(\theta, \phi)$, or use some form of a noninformative prior in the absence of any prior information on the parameter of interest. Informative priors should be possible in historical or pilot studies.

2. Fit the model $p(y|x, w, \theta)$ to the sample data; obtain posterior simulations $\theta^s, s = 1, \ldots, S$.

3. Given a prior distribution for ϕ, fit the model $p_{\text{sample}}(w|x, \phi)$ to the sample data; obtain posterior simulations $\phi^s, s = 1, \ldots, S$. If θ and ϕ share parameters or are dependent in their prior distribution, these two models would be fit together in one step.

4. For each draw (θ^s, ϕ^s):

   (a) For each poststratification cell j:

      i. Draw weights w^l, $l = 1, \ldots, L$, from $p_{\text{sample}}(w|X_j, \phi^s)$.

      ii. For each w^l, compute $\text{E}(y|X_j, w^l, \theta^s)$ from the regression model.

      iii. Compute the weighted average $\overline{Y}_j^s = \sum_{l=1}^L w^l \text{E}(y|X_j, w^l, \theta^s, \phi^s) / \sum_{l=1}^L w^l$, which is a Monte Carlo estimate of $\text{E}(y|X_j, \theta^s, \phi^s)$, the population mean within cell j under the model.

   (b) Compute the inferred population mean $\overline{Y}^s = \sum_{j=1}^J N_j \overline{Y}_j^s / \sum_{j=1}^J N_j$ and any subpopulation means or comparisons of interest.

5. Approximate the posterior distribution of all quantities saved in the previous step by their S simulations.

The workflow would then be continued with the usual steps of checking computational accuracy, model fit, sensitivity, and altering or expanding the model as necessary (Gelman et al., 2013, 2020).

This approach should automatically give stable small-area estimates, as long as the factors defining the small areas are included in x and a rich enough set of models is used to fit regressions (2) and (3). Indeed, this is the main contribution of our approach, that it seamlessly performs weighting adjustment within a modeling context that allows small-area estimation and poststratification.

The MRPW estimator $\widehat{Y}^{MRPW} = \frac{\sum_{j=1}^J N_j \sum_{l=1}^L w^l \text{E}(y|X_j, w^l, \theta, \phi) / \sum_{l=1}^L w^l}{\sum_{j=1}^J N_j}$ is a composite of the weighted and model-estimated cell mean estimates, which is further corrected by poststratification. Based on the design-consistency property of hierarchical Bayesian estimators examined by Lahiri and Mukherjee (2007), the MRPW estimator is design consistent if the weights and poststratification have sufficiently summarized the design features.

If there is interest in within-cell population summaries other than averages, then step 4(b) of the above algorithm must be made more general. Instead of simply computing a weighted average over the draws w^l, we can use Pareto-smoothed importance resampling to draw a subset $L^* < L$ of these weights with probabilities proportional to w^l (Vehtari et al., 2015). Collect the L^* resampled draws and renumber them as $w^{l^*}, l^* = 1, \ldots, L^*$. These approximate a set of draws from the population model, $p(w|X_j, \phi^s)$. For each w^{l^*}, we can then continue by sampling one value y from the predictive distribution, $p(y|X_j, w^{l^*}, \theta^s)$. We can then complete the process by computing whatever summaries are desired using the L^* draws of y within that cell (including regression coefficients).

3. Implementation

We consider how to efficiently apply our method for the problem of estimating summaries based on population cell means; as discussed in Section 2.2, this should suffice for most applications of survey research, including estimating the mean of the outcome y in the population or subsets of the population determined by the predictors x, along with population linear and logistic regressions of y on x.

3.1. Closed-form solution with a lognormal or gamma model for the weights

The algorithm just described has a cumbersome nested design requiring a new draw of $w^1, \ldots, w^L$ conditional on each posterior draw of the model parameters, along with a potentially unstable weighted averaging step.

One way to speed the computation is to use a model for the weights where the denominator of (4) can be evaluated in closed form. One such model, proposed by Skinner (1994), is lognormal regression.

Suppose we define $v = \log w$ and fit the model, $p_{\text{sample}}(v|x) = \text{normal}(v|g(x,\beta),\, \sigma)$, where $g(\cdot)$ is a function of x given parameter vector β, so that $\phi = (\beta, \sigma)$. Then (4) can be written as

$$p(v|x,\phi) = \frac{e^v p_{\text{sample}}(v|x,\phi)}{\int e^v p_{\text{sample}}(v|x,\phi)dv}, \tag{6}$$

and we can simplify the expression that appears in the numerator and denominator:

$$\begin{aligned} e^v p_{\text{sample}}(v|x,\phi) &= e^v \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(v-g)^2} \\ &= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}((v-g)^2 - 2\sigma^2 v)} \\ &= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}((v-(g+\sigma^2))^2 - \sigma^4 - 2g\sigma^2)} \\ &= e^{g+\frac{1}{2}\sigma^2}\, \text{normal}(v|g+\sigma^2, \sigma), \end{aligned}$$

so that (6) becomes,

$$\begin{aligned} p(v|x,\phi) &= \frac{e^{g+\frac{1}{2}\sigma^2}\, \text{normal}(v|g+\sigma^2,\sigma)}{e^{g+\frac{1}{2}\sigma^2}} \\ &= \text{normal}(v|g+\sigma^2,\sigma). \end{aligned} \tag{7}$$

Thus, under the lognormal model, the population distribution of the weights is identical to the sample distribution except that it is shifted to the right by σ^2. This makes sense. First, a large weight corresponds to more representation in the population, so we should expect higher weights to be more common in the population than in the sample. Second, σ^2 is the residual variance of the log weights, so the higher the value of σ, the more consequential will be the weighting (in terms of shifting estimates, if the weights are correlated with the variable of interest, or in terms of reducing precision, if the weights are independent of the variable of interest), hence the larger the shift. At the extreme of $\sigma = 0$, the weights do not vary within poststratification cells at all, and no adjustment is needed.

The above calculation took advantage of a conjugacy property of e^v with the normal density. Closed-form computation is available under other models as well. For example, if the weights follow a gamma regression, then multiplying the density function by w has the effect of adding 1 to the shape parameter α of the model and correspondingly shifting the mean upward, so that if the mean of the distribution of $p_{\text{sample}}(w|x,\phi)$ in the sample is g, then the mean in the population distribution $p(w|x,\phi)$ becomes $\frac{\alpha+1}{\alpha}g$, which again makes sense, both in that it is an increase compared to the sample and that the increase goes to zero in the limit of $\alpha \to \infty$.

3.2. Closed-form solution with a mixture of lognormals or gammas

For various reasons, the distribution of weights can be far from normal or gamma. But we can retain the clean computation of these conjugate forms using a mixture model. We demonstrate with the lognormal.

Start with the model, $p_{\text{sample}}(v|x,\phi) = \sum_{m=1}^{M} \lambda_m \,\text{normal}(v|g_m(x,\beta),\sigma_m)$, where g_m is a family of regression functions given parameter vector β, so that $\phi = (\beta, \lambda_m, \sigma_m)$. Similar algebra as before yields the population distribution,

$$p(v|x,\phi) = \frac{\sum_{m=1}^{M} \lambda_m \, e^{g_m(x,\beta)+\frac{1}{2}\sigma_m^2} \,\text{normal}(v \mid g_m(x,\beta)+\sigma_m^2, \sigma_m)}{\sum_{m=1}^{M} \lambda_m \, e^{g_n(x,\beta)+\frac{1}{2}\sigma_m^2}}, \tag{8}$$

which again is a mixture of M lognormals. In addition to each mean being shifted by σ_m^2 as before, the mixture proportions change, with components with higher values counting more in the population, which makes sense. For the poststratification, we will need to compute the mixture components in (8) for each poststratification cell, using the predictors X_j.

In practice it may be enough to model the weights using a lognormal or gamma error term or perhaps mixtures of one of these. But if a more general model is desired, it should be possible to get much of the computational benefits by first fitting the closed-form model and then using it as an approximation for the desired model. Instead of importance ratios w^l in step 4(b) on page 7, one would use the ratio of the exact and approximate densities, which should be more stable.

3.3. Allowing the distribution of weighting-model residuals to vary across poststratification cells

In our procedure, the adjustment for unequal-probability sampling is performed by reweighting the distribution of log weights v conditional on predictors x. Sections 3.1 and 3.2 considered models where the distribution of $v|x$ has a common form across postratification cells, with mean determined by a regression model and variance estimated based on the residuals from the regression fit to all the data. The variance of the residuals then determines the amount that the estimated distribution of weights need to be shifted upward to account for unequal-probability sampling.

But what if the variance of the weights itself varies across poststratification cells? In that case it is not appropriate to shift the log weights in all cells by the same amount; such a procedure can lead to a biased estimate.

We demonstrate the problem using a simple hypothetical example of a survey with only two poststratification cells, women and men, coded as $x = 0$ and 1, respectively. We assume that women have been oversampled and comprise two-thirds of the sample. We further assume that the weights depend on various unobserved factors that vary more with men than with women, so that the log weights are higher and more variable for men than for women, following these distributions: $p_{\text{sample}}(v|x=0) = \text{normal}(v|0, 0.5)$, $p_{\text{sample}}(v|x=1) = \text{normal}(v|0.5, 0.8)$. We have set these values so that the average weight for men in the sample is approximately twice that of the average weight of women: $e^{0+\frac{1}{2}(0.5)^2} = 1.13$; $e^{0.5+\frac{1}{2}(0.8)^2} = 2.27$.

To get the population distribution of $v|x$, we simply follow equation (7) and shift upward by the residual variance; thus, $p(v|x=0) = \text{normal}(v|0.5^2, 0.5) = \text{normal}(v|0.25, 0.5)$, $p(v|x=1) = \text{normal}(v|0.5+0.8^2, 0.8) = \text{normal}(v|1.14, 0.8)$.

Unfortunately, this is not the estimated distribution of $v|x$ obtained by fitting a regression model with pooled variance. In this case, with only a single predictor, the regression picks up the sample mean of v within each sex and a pooled sample variance, which will be estimated as approximately

$\frac{2}{3}(0.5)^2 + \frac{1}{3}(0.8)^2 = 0.38$, thus an estimated residual standard deviation σ of $\sqrt{0.38} = 0.62$, and so the inferred (but incorrect) population distribution of v is normal$(0.38, 0.62)$ for women and normal$(0.88, 0.62)$ for men.

These estimated distributions of log weights for the two sexes differ from the true population distributions, even in the limit of large amounts of data. And this can have a devastating impact on inferences for any outcome y that is correlated with v. Suppose, for example, that y is height in centimeters, x is an indicator for being male, and the data are well fit by a model, $y = 161 + 6x + 7xv +$ error. We have set up a scenario in which, conditional on sex, taller men (but not taller women) are less likely to be included in the sample (on average, they have higher weights) and have set up the numbers so that the average heights for women and men approximately correspond to known population averages: given the numbers above, $\mathrm{E}(y|x=0) = 161$ and $\mathrm{E}(y|x=1) = 161 + 6 + 7 \cdot 1.14 = 175$. If instead we use the estimated population distribution of v from the regression model with pooled variance, we get the wrong answer of $\mathrm{E}(y|x=1) = 161 + 6 + 7 \cdot 0.62 = 171.3$. By using the pooled error distribution for v in this example, we greatly underestimated the variance of weights among men, which in turn reduced the shift when adjusting from sample to population distribution of $\mathrm{E}(v)$ in (7). What is distressing is that the error does not go away as sample size increases; that is, the estimate is inconsistent.

We can address this problem by setting up the model for $v|x$ so that the variance as well as the mean varies with x, following ideas used by Maiti et al. (2014), Sugasawa et al. (2017), and Savitsky et al. (2022) for small-area estimation and adapting them to the weighting problem. The simplest approach is to extend the normal model of Section 3.1 as follows:

$$p_{\text{sample}}(v|x, \phi) = \text{normal}(v|g(x, \beta), h(x, \gamma)), \tag{9}$$

with separate regression models for the mean and variance, with parameter vectors β and γ estimated from the data, and $\phi = (\beta, \gamma)$. A natural choice of parametric form would be linear on the location and loglinear on the scale:

$$p_{\text{sample}}(v|x) = \text{normal}(v|x\beta, e^{x\gamma}),$$

In any case, the distribution of the log weights would be shifted by the variances, as with (7); thus (9) yields

$$p(v|x, \phi) = \text{normal}(v|g(x, \beta) + h(x, \gamma)^2, h(x, \gamma)). \tag{10}$$

One could similarly alter the gamma and mixture models as well.

3.4. Weighted bootstrap of regression residuals

An alternative to modeling the distribution of regression residuals is to bootstrap them, resampling in proportion to the weights (Bertail and Combris, 1997; Cohen, 1997).

First consider the basic model with a shared error distribution across cells. We can then apply a weighted bootstrap to the full set of n residuals. If the residuals from the regression of v on x are $r_1, \ldots, r_n$, then for each poststratification cell j we sample L residuals with replacement from $\{r_1, \ldots, r_n\}$, with probabilities proportional to $\exp(r_i)$.

If the residuals average to zero, then, from Jensen's inequality, the distribution of residuals weighted by their exponentials has positive expectation. This makes sense: the expected value of the log weights is greater for the population distribution than for the sampling distribution (unless all weights are equal), and this should hold within each cell.

If the model for $\mathrm{E}(y|x,v,\theta)$ is linear in v, we can then proceed simply by computing the mean of the distribution of residuals of the weight model, that is, $\bar{r}_{\text{weighted}} = \sum_{i=1}^{n}(r_i \exp(r_i))/\sum_{i=1}^{n}\exp(r_i)$, hence we just plug in $\mathrm{E}(v|X_j,\phi)+\bar{r}_{\text{weighted}}$ instead of $\mathrm{E}(v|X_j,\phi)$ for v_j when estimating $\mathrm{E}(y|x_i,v_j,\theta)$ within each poststratification cell.

If $\mathrm{E}(y|x,v,\theta)$ is nonlinear in v, as with logistic regression, then we would want to estimate the expectation averaging over v, $\mathrm{E}(y|x,\theta)$, using sampling, for each cell j imputing values for $v|X_j$ by taking the predicted value for the cell, $\mathrm{E}(v|X_j,\phi)$, and then adding random draws from the residuals sampled using the above-described weighted bootstrap.

3.5. Weighted bootstrap allowing the distribution of residuals to vary across cells

Next consider the more general case in which the distribution of the error of the model for $\mathrm{E}(v|x,\phi)$ varies by x. Bootstrapping over the entire distribution of residuals would then give the wrong answer, as explained in Section 3.3, but it would also in general not work to simply bootstrap the residuals within each cell. For poststratification cells with no data at all, there would be no residuals to bootstrap—and it would be wrong to set the imputed residuals to zero for such cells, as this would underestimate the population cell mean, $\mathrm{E}(v|X_j,\phi)$. And if a cell has only a few data points, then it would have few residuals to booststrap, and the resulting adjustment would be noisy.

So it makes sense to compromise between bootstrapping of within-cell residuals and bootstrapping of all the residuals in the data. For each cell j, this can be done using a weighted bootstrap of the residuals $\{r_1,\ldots,r_n\}$ from the fitted regression of v on x, using the following rule:

$$\text{bootstrap weight for } r_i\text{:}\ \ a_{ij}\exp(r_i), \tag{11}$$

where a_{ij} is some measure of closeness between observation i and cell j.

The simplest nontrivial version of (11) would set a_{ij} to a larger value if observation i is in cell j. The trick is to set this constant in such a way that the procedure gives stable answers for small cells while giving consistent estimates as the sample size n approaches infinity. For example, if we set $a_{ij} = 1/30$ for the data points i within cell j and set $a_{ij} = 1/(n-n_j)$ otherwise, then in a cell with $n_j = 30$, the bootstrap will give roughly equal total weight to the residuals within and outside the cell, cells with much fewer than 30 observations will mostly rely on the full sample of residuals, and cells with much more than 30 observations will mostly rely on the residuals within the cell. As n increases, all the cell sizes increase in expectation; thus, in the asymptotic limit, each cell's bootstrap is determined by the residuals within the cell, so that this part of the inference is consistent.

That simple reweighting is an incomplete solution in that it is a crude mix of hyperlocal (residuals within the single cell) and global (the entire sample). Ideally we would want a procedure closer to what is done in multilevel modeling, giving higher weights to residuals from cells that are close in the space of x.

As a generic choice, we could weight based on a Manhattan distance, which if all k predictors in x are discrete would be defined as,

$$d_{ij} = \sum_{k=1}^{K} 1_{x_{ik}\neq X_{jk}}, \tag{12}$$

and for any continuous predictors k, the term $1_{x_{ik}\neq X_{jk}}$ would be replaced by $|x_{ik}-X_{jk}|/S_k$, where S_k is set to 2 times the standard deviation of X_k (Gelman, 2008). The expression (12) is then a sum of the discrepancies across the K predictors.

We then can then define the bootstrap weights using a formula such as,

$$\text{bootstrap weight for } r_i\text{: } \begin{cases} (1/30)\exp(r_i) & \text{if } i \text{ is within cell } j \\ (c_j/d_{ij})\exp(r_i) & \text{otherwise,} \end{cases} \tag{13}$$

where $c_j = (\sum_{i;\, d_{ij}>0} \frac{1}{d_{ij}})^{-1}$. Expression (13) assigns greater weight to residuals from nearby cells and is consistent in the limit of large n_j, with roughly equal weighting of within-cell and out-of-cell residuals when $n_j = 30$.

3.6. Integrated Bayesian computation

We can perform all the steps of regression and poststratification in a single probabilistic program when performing the weighting adjustment using a closed-form solution or bootstrap simulation. The computation goes as follows:

1. Specify models $p_{\text{sample}}(v|x,\phi)$ and $p(y|x,v,\theta)$ and estimate ϕ and θ together. Joint estimation of the two models would not, strictly speaking, be necessary if the parameter vectors ϕ and θ are distinct and independent in their prior distribution, but it is convenient to perform inference within the same probabilistic program so that we can work with posterior simulations from both of them together in the next step.

2. Loop through the poststratification cells $j = 1, \ldots, J$: for each cell j, sample L draws v^l from the estimated or approximated $p(v|X_j,\phi)$ using the closed-form solution or weighted bootstrap. Propagate each simulated v^l through the regression model for the outcome variable to compute $\mathrm{E}(y|X_j,v^l,\theta)$, and then average over these to obtain a Monte Carlo estimate of $\mathrm{E}(y|X_j,\theta)$.

3. The result of the above steps is an $S \times J$ matrix of simulations representing the posterior distribution of the population mean in the J poststratification cells; these can be combined to get posterior estimates and uncertainties for the poststratified population mean or any subset of the population defined in terms of the predictors x.

3.7. Large number of poststratification cells

In MRP, it is common to have many poststratification factors, leading to a large number of cells. If each factor $k = 1, \ldots, K$ has J_k levels, then there are $J = \prod_{k=1}^{K}$ poststrafication cells, a number that can quickly become very large, possibly much than the sample size n of the survey, so that most cells are empty or only contain one or two people in the sample.

With multilevel modeling, this sparsity does not pose a problem for inference, but it can cause computational challenges, especially for the bootstrapping approach, which requires a resample of L values of $v|x_j, phi$ for each cell j. This requires potentially three levels of looping, corresponding to the L bootstrap samples, the J cells, and the S simulation draws of ϕ.

But if the number of cells is huge and no single cell represents a large chunk of the population, it should be enough to take a single bootstrap draw of v within each cell, that is, to set $L = 1$. The result should still yield acceptable population inferences for regressions and small areas of interest, because the aggregation over all the tiny cells performs an average over the distribution of (v, x), even with just one draw of v per cell. As we show in Appendix A, this allows us to simplify our computations for large multilevel models and avoid an awkward additional level of looping.

3.8. Continuous predictors and the poststratification file

When the predictors x are continuous, or when they are discrete but the number of poststratification cells J is large compared to the population N, it can make more sense to talk about a poststratification *file*, which is simply a matrix with N rows giving the values of the vector x for everyone in the population and is thus a special case of the poststratification table in which $N_j = 1$ for all j. An example is political analysis using the voter file (Ghitza and Gelman, 2020). The practical relevance is that each entry in the file represents just one person in the population, and thus when simulating or boostrapping $v|x$, a single draw v_j is needed for each population item j, thus $L = 1$ in Section 3.6, which removes one loop from the computation. In addition, we would then use the continuous-predictor distance function in (12).

4. Understanding through limiting cases

To explore MRPW, we consider various special cases in which different information is available to the practitioner.

4.1. Weights that entirely depend on the background variables

We first consider the case in which w is a deterministic function of x. In this case, the regression $p_{\text{sample}}(v|x)$ will fit perfectly; that is, the residuals will all be zero, and $p(v|x) = p_{\text{sample}}(v|x)$, and the resulting inference, based on separately fitting $p(y|x, v)$ and $p(v|x)$, will be essentially the same as that obtained by just fitting $p(y|x)$ directly and bypassing v entirely. Our procedure thus reduces to the standard advice to ignore the sampling weights if the inclusion probabilities are entirely determined by x.

4.2. No background variables

Now suppose the analyst only knows y and w and is not given x at all. As always, the subsequent analysis will depend on the model that is fit.

The simplest parametric approach uses a normal model for log weights v and linear regression for $y|v$; thus, $v \sim \text{normal}(\mu_v, \sigma_v)$ and $y|v \sim \text{normal}(a+bv, \sigma_y)$. Conditional on the estimated model parameters, the population mean is then

$$\overline{Y} = \text{E}(y) = a + b\bar{v} + b\sigma_v^2. \tag{14}$$

This is the prediction from the sample model adjusted to increase the representation of data points with higher weights. The point estimate would be

$$\widehat{\overline{Y}} = \hat{a} + \hat{b}\bar{v} + \hat{b}\sigma_v^2. \tag{15}$$

Assuming the regression of y on v has been estimated by least squares, (15) becomes,

$$\widehat{\overline{Y}} = \bar{y} + \hat{b}\sigma_v^2. \tag{16}$$

Expression (16) makes sense—it is the familiar regression estimate, $\bar{y} + \hat{b}\,(\overline{V} - \bar{v})$ from sampling theory, which linearly adjusts for differences in the average of the control variable in the sample and its average in the population. In our model, $\overline{V}$ is not known but is inferred from the observed distribution of the weights. Finally, this gives us a point estimate. We can obtain a posterior

distribution for $\overline{Y}$ from (14) by averaging over the inferential uncertainty in (a, b, σ_v) from the two fitted regressions.

Alternatively, we can analyze the bootstrap procedure, which is simple here: with no predictors x, there is effectively only one poststratification cell, so the implied population distribution of the weights takes on the values of the observed w_i's, each with probability proportional to $w_i = e^{v_i}$. The population mean is then,

$$\overline{Y} = \mathrm{E}(y) = \frac{\sum_{i=1}^n e^{v_i} \mathrm{E}(y|v_i)}{\sum_{i=1}^n e^{v_i}}. \tag{17}$$

In this case, the bootstrap simulation is not required as we can compute the expectation over the bootstrap distribution directly.

As before, inference for $\overline{Y}$ will depend on the model for $\mathrm{E}(y|v)$. If we again consider a default normal model, $y|v \sim \text{normal}(a + bv, \sigma_y)$, the population mean is then,

$$\overline{Y} = \mathrm{E}(y) = a + b\frac{\sum_{i=1}^n e^{v_i} v_i}{\sum_{i=1}^n e^{v_i}}, \tag{18}$$

If a and b are estimated by least squares, this results in a point estimate,

$$\widehat{\overline{Y}} = \bar{y} + \hat{b}\left(\frac{\sum_{i=1}^n e^{v_i} v_i}{\sum_{i=1}^n e^{v_i}} - \bar{v}\right), \tag{19}$$

which again can be framed as the regression estimate, $\bar{y} + \hat{b}\,(\overline{V} - \bar{v})$, just with a nonparametric estimate of $\overline{V}$.

4.3. Other scenarios

One way to better understand how MRPW works and where it breaks down is to consider other simple scenarios using simulated data. For example:

- Simple stratified sampling with weights; then perform the analysis ignoring the strata. How does this differ from the standard stratified analysis? How does it differ from a weighted-average analysis ignoring the strata?
- Finite-population sampling including a certainty stratum and thus a hard lower bound on weights.
- Poststratification weights modeled as inverse-probability weights: how much does our approach inflate the variance estimate compared to the correct poststratification analysis?
- Simple small-area estimation without or with a group-level predictor. Result will depend on the dependence between weights and expected outcome, so try different possibilities in the simulations.
- Small-area estimation with a huge number of cells so $n_j = 0$ or 1 in almost all cells and there is no observable variation in weights within each cell, but the weights still matter.
- Weights that have been estimated for non-probability samples based on quasi-randomization or doubly robust approaches (Chen et al., 2020). How well does this procedure perform in this case in which the weights are only estimates of inverse probabilities of selection?

- No available poststratification information on the background variables. In this case, the distribution of X needs to be estimated using available data. One way to address this is to poststratify on the data—that is, to use the observed data x as the postratification file X.

These scenarios raise conceptual challenges. Consider, for example, a national survey that is poststratified by geography, in such small areas (for example, zip codes) that there are no cells with more than one respondent in the sample. Also suppose that the survey weights are not based on geography but are instead based on the number of people living in the respondent's household, a variable that is not otherwise included in the analysis. The weights will still vary by geography even though they are not defined explicitly in geographic terms. But with only one respondent per poststratification cell in the data we cannot estimate the within-cell variance in log weights (the crucial parameter σ^2 in model (7)), and the only way forward, short of including the "number of people in the household" variable in the analysis, might be to combine cells to allow the estimation of within-cell variation of the weights.

Add some structure to the problem, though, and it becomes easier to solve. Take the same example, with the same number of poststratification cells, but suppose they are formed by the intersection of several variables, for example age, sex, ethnicity, education, and congressional district. In this case, a multilevel regression of log weight on these factors will yield a nonzero residual variance, as long as the model does not include the fully-saturated interaction of all the predictors.

5. Simulated-data examples

5.1. Logistic regression for a hypothetical marketing study

One can develop understanding of the method using simulated datasets for which the sampling model (1) is correct, so that the survey weights w_i are truly proportional to the inverse probabilities of inclusion in the sample.

We first demonstrate our method with a hypothetical marketing study. The population comprises the N users who visit a webpage during a month, a small fraction of whom are given the opportunity to buy a certain product. Each user i has a customer engagement score, x_i, and the probability π_i of user i being exposed to the promotion is determined by an algorithm that depends on x_i and some other factors. The distribution of x_i is known in the population, and we know the values of π_i for the n users in the sample. Label y_i as the binary outcome of whether person i in the sample buys the product. The goal is to estimate the proportion of people in the population who would buy the product, and also to learn how this probability varies with x.

We simulate data based on the following assumptions:

1. $N = 10^6$.

2. The engagement scores x_i in the population are uniformly distributed in the set $\{1, 2, \ldots, 10\}$.

3. The probability of inclusion in the sample is an increasing function of engagement score, but with variation: $\pi_i = 10^{-4} x_i \exp(\epsilon_i)$, where $\epsilon_i \sim \text{normal}(0, 0.8)$.

4. The probability of buying the product is an increasing function of the engagement score; beyond that, it increases with the probability of being in the sample, following the rule, $\Pr(y_i = 1) = \text{logit}^{-1}(0.9 + 0.1x_i + 0.5 \log \pi_i)$. We set the coefficients for x_i and π_i so that both would be relevant to the outcome, and then we set the intercept so that the buy rate in the sample is approximately 10%.

5. The survey is conducted, and the analyst is given the distribution of x in the population and the following information on the n respondents in the sample: the engagement score x, the response y, and a weight w which is proportional to $1/\pi$. The analyst is not given the values of π in the population.

In Appendix A, we provide code and we show how implementing the methods we have introduced in Section 3 recovers the population parameters in this example.

5.2. Constructing simulated-data examples from real surveys

Simulation studies can also be conducted by subsampling from real survey data. In that case, the "population" (a large existing survey) is completely known, we have full control over the sampling procedure, we can define weights however we want, and we can compare our inferences to the population values, checking accuracy of estimates and coverage of uncertainty intervals.

Suppose for example we start with the poststratification table based on U.S. Census data for 50 states $\times$ 4 ethnicity categories $\times$ 5 education categories $\times$ 6 age categories $\times$ 2 sexes. In addition, we construct an additional variable z that takes on the values $\{1, 2, \ldots, 10\}$ and is uniformly distributed in the population independently of the poststratification variables. It is assumed that z is known to the group that conducts the survey and thus they can use it when creating sampling weights, but it will not be available to later analysts. In that way, z stands for all the information that is used to construct weights but is not recorded for users.

We could assume a sampling model such as,

$$\Pr(\text{unit } i \text{ is included in the sample}) \propto h(x_i) z_i^{\eta}, \tag{20}$$

where h is a function constructed ahead of time by fitting a hierarchical logistic regression model to the respondents of a large real survey such as the Cooperative Congressional Election Study used in Section 6, predicting probability of response from the above-listed poststratification factors. The exponent η is a tuning parameter that we can set to different values to modulate the importance of z in the weighting. If $\eta = 0$, the weights depend entirely on the poststratification variables x, as discussed in Section 4.1; If $\eta = 1$, these weights strongly depend on the unobserved z.

We can then sample n people from the assumed distribution with probabilities proportional to those given above. For each respondent i we would record a weight w_i that is proportional to the inverse of the sampling probability.

We could then simulate a discrete survey response, starting by fitting a multilevel regression predicting a survey response given the above-listed demographic and geographic predictors, to obtain a fitted model with vector of coefficients $\hat{\beta}$ and residual standard deviation $\hat{\sigma}$). We could then simulate data from the model $y \sim \text{normal}(x\hat{\beta} + 0.2\,(z - 5.5)\,\zeta,\ \hat{\sigma})$, thus inducing a large sampling bias. Recall from the above that z is uniformly distributed on $\{1, 2, \ldots, 10\}$ in the population, but inclusion in the sample can depend on z.

We would complete the simulation by constructing a dataset with y, w, $v = \log(w)$, and the variables in x. In all our scenarios, the goal would be to estimate the population average, $\overline{Y}$, and we would proceed as follows:

1. Fit a multilevel linear regression for $p(y|x, v)$, obtaining posterior simulation draws for the parameters in this fitted model.

2. Fit a multilevel linear regression for $p_{\text{sample}}(v|x)$, obtaining posterior simulation draws for the parameters in this fitted model.

3. For each of S posterior simulation draws:

   (a) Within each poststratification cell:

      i. Perform the weighted bootstrap of residuals as described in Section 3.5 and add these to the fitted regression to obtain an estimated population distribution $p(v|x)$ given that cell.

      ii. Plug these simulations of v into the fitted model from Step 1 above to estimate $\mathrm{E}(y|x, v)$ within the cell.

   (b) Poststratify to obtain an estimate the population mean, $\overline{Y}$.

4. Take these S estimates as the posterior distribution for $\overline{Y}$.

6. Real-data example

Evaluating an adjustment method in applied examples can be difficult because for most survey questions we do not know the true population values. One setting where we do know the truth, and which we have used to evaluate MRP in the past, is U.S. election polling; however, challenges arise there too given problems of differential nonresponse (Little and Gelman, 1998; Brick and Tourangeau, 2017; Kuriwaki et al., 2024).

That said, we still think much can be learned by applying a new procedure to real problems. The method can run into computational difficulties, it can give unreasonable results, and the model could have problems fitting to data. More positively, we can get a sense of distributions of weights in real surveys and compare different approaches to small-area estimation and regression modeling in the presence of survey weights.

Here we demonstrate the use of MRPW to obtain state-level estimates of attitudes on six abortion-related policies from a national poll, the 2018 Cooperative Congressional Election Study (CCES), which we choose because it has open data and has been used earlier for an MRP case study (Lopez-Martin et al., 2022). We poststratify on ethnicity, age, education, and state. We work with a random sample of 500 respondents so as to highlight the effects of multilevel modeling for small-area estimation. In addition to the poststratification variables, the model includes two state-level predictors: an indicator for region and the Republican vote share in the state in the 2016 presidential election.

We perform two studies, one using linear regression and one using logistic regression. For each study, we fit a series of MRP models with different numbers of predictors, thus moving along the spectrum between the two extremes of having many pre-treatment predictors (so that the weights little within each poststratification cell and thus do not contain much additional information) to having no pre-treament predictors so that all the adjustment information is carried by the weights.

6.1. Multilevel linear regression

For our first study, we use a composite response on a 0–6 scale (which we treat as a continuous outcome), with higher values representing more support for abortion.

We use the survey weights that come with the CCES, which include demographic information but are not simply determined by the predictors in our model, as can be seen from the fitted multilevel regression for the log weights v given poststratification variables x:

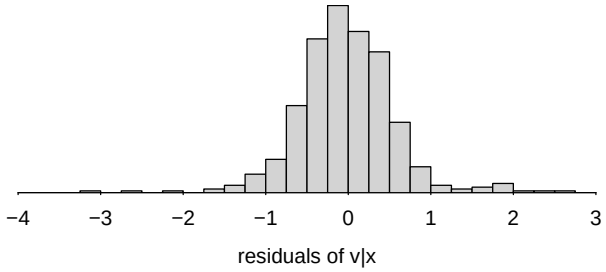


Figure 1: *Histogram of residuals of the regression of log weights for the CCES example. The distribution has wider-than-normal tails.*

```
            Median MAD_SD
(Intercept) -0.1    0.3
male         0.4    0.2
repvote     -0.1    0.0

Auxiliary parameter(s):
      Median MAD_SD
sigma 0.6    0.0

Error terms:
 Groups   Name        Std.Dev.
 state    (Intercept) 0.135
 educ:age (Intercept) 0.110
 educ:eth (Intercept) 0.305
 male:eth (Intercept) 0.252
 age      (Intercept) 0.078
 educ     (Intercept) 0.633
 region   (Intercept) 0.167
 eth      (Intercept) 0.265
 Residual             0.609
```

Figure 1 displays a histogram of the residuals from the regression of log weights. Recall that here we are working with a random sample of 500 survey respondents. When looking at the full survey with data from 60,000 people, the distribution of weights has another mode, corresponding to less than 1% of the data, of very small weights, less than -8 on the log scale. These data points will be negligible in any weighted analysis; however, because of their distance from the large mass of the data, they would have some influence on the coefficients for v in the model of $y|x, v$, and we would recommend removing these cases with extremely low weights before proceeding.

Figure 2 shows the estimated average attitudes for the 50 states based on unweighted MRP, weighted-likelihood MRP, and our recommended MRPW approach. In this case, the results from the power-weighted approach seem off, with the states in the western region being estimated to be much more liberal in abortion attitudes compared to the rest of country. MRPW gives similar results to unweighted MRP in this example, which makes sense—in practice, weighting often has a small impact on final results. The benefit of including weights is to address general concerns of non-representativeness of the sample. It is not necessary in our example for MRPW to give much different answers than unweighted MRP; rather, the benefit is that we can incorporate weights into MRP, thus not needing to choose between the two approaches, and avoiding the problems with

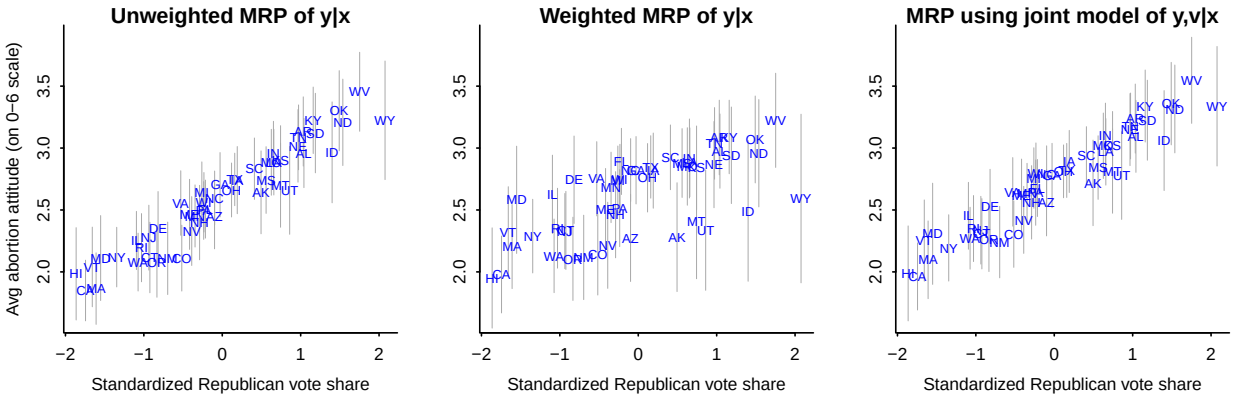


Figure 2: *Posterior estimates* ± 1 *standard deviation for state-level opinion averages with a continuous outcome based on three different multilevel regression and poststratification (MRP) analyses fit to a sample of 500 respondents: (a) MRP applied to the unweighted data, (b) MRP using the weights as powers of the likelihood factors, (c) our recommended MRPW approach using a joint model for weights and outcomes.*

weighted likelihoods.

The coding details for this analysis appear in Appendix A.

6.2. Multilevel logistic regression

We repeat the above study with a series of multilevel logistic regressions for a binary response, using just one of the six abortion questions in the CCES. The analysis of the log weights remains unchanged, with the same distribution of residuals as in Figure 1. Figure 3 shows the inference for the population proportions by state, compared to MRP and weighted MRP of the data. As with the continuous-data models in Figure 2, the joint model gives similar results to the unweighted analysis. More generally, though, the weights can make a difference, in which case we want to apply them appropriately.

7. Concerns

7.1. Unrealistic assumptions of the model

We call our method quasi rather than fully Bayesian because it is based on a generative model in which the weights w are defined in the population and are drawn to create the sample, but in real surveys the weights are constructed and defined only for the sample, and they have no underlying population distribution. In that way, our approach is similar to many applications of statistics in which a probability model is used even in the absence of any superpopulation or physical randomization (Little, 2004; Elliott and Valliant, 2017).

Our model assumes independent sampling with probabilities proportional to $1/w$, but, as noted by Gelman (2007), survey weights are often constructed by raking and do not represent sampling probabilities at all. Even when weights are intended to represent inverse sampling probabilities, they generally do not, as the construction of weights is only approximate.

Why would we purposely construct a model that is wrong in these crucial ways? The short answer is that, to the extent that weights are well constructed in a practical sense and used as intended, an item with weight w in the sample is intended to represent w items in the population.

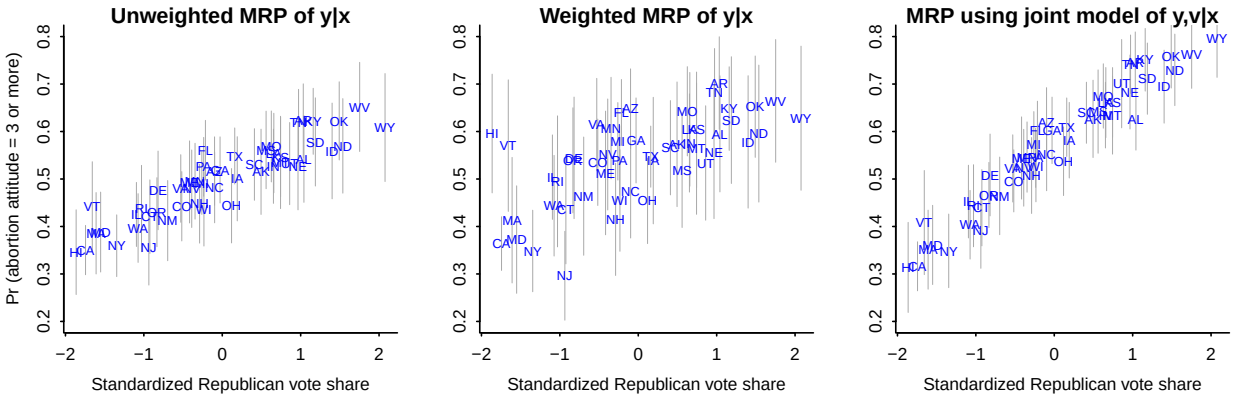


Figure 3: *Posterior estimates* ± 1 *standard deviation for state-level opinion for a binary outcome based on three different multilevel regression and poststratification (MRP) analyses fit to a sample of 500 respondents: (a) MRP applied to the unweighted data, (b) MRP using the weights as powers of the likelihood factors, (c) our recommended MRPW approach using a joint model for weights and outcomes.*

Our procedure can be viewed as a smoothed version of applying weights to items in the sample. To the extent that we are building a model-based adaptation or generalization of existing practice, it makes sense to take the weights seriously and consider them as being inversely proportional to the probability of inclusion in the sample, even if they are not. Similarly, the assumption of independent sampling can be viewed as an instantiation of the recommended methods in which weights are attached to individual units. As we have written elsewhere, we fit a model consistent with standard practice because we want our approach to be an improvement upon rather than merely a replacement for standard weighted analysis of sample surveys. Similar ideas can be applied using non-Bayesian methods (Morikawa et al., 2022).

Another potential concern is the use of a regression, $p(w|x)$, that implies a continuous distribution of weights in the population, even though weights in real surveys typically take on only a finite possible number of values. In the past we have considered nonparametric modeling of survey weights (Si et al., 2015), but this adds enough complexity to the analysis that we have avoided it here. In practice, we think the lognormal or lognormal-mixture regression model used in the present paper should be fine: the lognormal regression should be a reasonable fit to weights that are constructed by multiplying many individual factors, mixture modeling can capture the discreteness that can arise if weights are dominated by one or two factors, and the weighted bootstrapping of residuals of regression for the log-weights should account for the rest. Also, to the extent that the weights depend on variables in x, much of their variation will be explained by the deterministic part of the weighting regression anyway.

7.2. Sensitivity to large weights

As with weighting-based methods in general, we need to be concerned about the right tail of the weight distribution, for two reasons. First, survey weights are often smoothed or trimmed to reduce their variability, which can make sense as a variance-reduction tool but complicates their interpretation. Second, large weights correspond to lower probabilities of sampling or survey response, so they represent "dark matter" in the sampling procedure: potentially large chunks of the population that are expected to appear rarely or not at all in the data. Any resolution

of this problem requires strong assumptions, such as a hard cap on the maximum weight in the population or a short upper tail that limits the total proportion of the population that would have large weights. In a finite-population analysis there is also a bound on the low end, because the probability of inclusion in the sample can never exceed 1.

One advantage of the bootstrap-the-residuals procedure described in Section 3.5 is that this automatically bounds the weights. Data resampling cannot go beyond the range of the data being sampled, and this limited capacity for generalization has been pointed out as a weakness of the bootstrap for statistical inference problems (Schenker, 1985). With sampling weights, however, this bounding of the bootstrap sampling is a strength, not a weakness, in that it shuts off the otherwise theoretically unmanageable upper tail of the distribution of weights.

When considering various aspects of sensitivity to model assumptions, remember that the goal is to estimate the population regression function, $p(y|x)$; the weights are just a means to this end, a way of adjusting for the biases that would occur if one were to attempt to extrapolate from a fitted model without adjusting for known discrepancies between sample and population. What is relevant, then, is the dependence of $p(y|x, w)$ on w. If this model is a smooth function of w, then approximating a discrete distribution of w by a continuous distribution might not cause serious problems. If the model behaves calmly for large values of w, then the "dark matter" problem of very large weights in the population might not be such a concern. It should be possible to do some theoretical analysis, looking at the tails of the model for w along with the functional form of $p(y|x, w)$ for large w to ensure bounded influence from the unobserved items with large weights. Working with deciles of the actual weight values may also be helpful for this kind of sensitivity analysis.

7.3. Weights that are negative, zero, or positive but very small

A dataset can include observations with zero weights, which is a signal to exclude them from analysis entirely. In our procedure it makes sense to remove these data points before beginning the analysis to avoid the awkward and otherwise unnecessary step of modeling a weight variable that can be zero or positive.

There are settings where negative weights can make sense as part of a regression adjustment (Ben-Michael et al., 2023), but these cannot be interpreted as inverse probability of inclusion in the sample, and so negative weights cannot fit into the methods used in this paper. In such settings, we would either restrict to data with positive weights or try to remove the steps within the weighting process that produced negative weights.

Finally, data points with extremely tiny weights will have essentially no effect on any weighted averages, but they can interfere with our estimation procedure by driving up the estimated variance of the weights in the population. One solution here is to fit a mixture model in which one of the components captures the low weights, thus effectively "quarantining" them so as not to contaminate inferences for population averages. It can also make sense to apply a simpler approach and just exclude such extreme cases.

8. Conclusion

The problem being attacked in this paper is to model an outcome y given predictors x from a sample whose data are collected with unequal probabilities, and also given inverse-probability weights w. We assume the joint distribution of x in the population is known. Our approach is to first estimate the w given x in the sample, then adjust (using the assumption of inverse-probability weighting) to

estimate the distribution of w given x in the population. We then fit a model predicting y given x and w. The process is completed by averaging that fitted model over w to obtain the desired goal of an estimated distribution of y given x. This can then be averaged over the population distribution of x ("poststratification") to obtain inferences for the entire population or for subsets defined by x.

MRPW unifies existing design-based weighting methods and model-based approaches to small-area estimation. Our approach goes beyond existing weighting methods by using a regression model for w given x, which allows us to escape the trap of what to do in small cells with only one or two observations. More generally, we can think of our procedure as a model-based adjustment for unequal sampling probabilities which plays well with MRP and other model-based approaches to small-area estimation.

Future challenges include inference with known margins (poststratification with marginal or lower-dimensional joint distributions, for example post-election adjustments based on local vote totals; Si and Zhou, 2021, Rosenman et al., 2023); cluster sampling (Makela et al., 2018); inference for non-census variables (for example, religion); and generalization from sample to population in causal inference (Miratrix et al., 2013; O'Muircheartaigh and Hedges, 2014; Kennedy and Gelman, 2021).

For now, our practical recommendations depend on where you stand in the process of data preparation and analysis:

- If you have access to the raw data and relevant population information: We would not typically recommend the methods in this paper. Instead of creating weights and then incorporating them into the analysis, it should be better to just model the data directly conditional on all information that might go into weighting. This might require augmenting the poststratification table (if there are relevant non-census variables: information predictive of the outcome and predictive of inclusion in the sample that is not available at the population level), but that modeling could be done directly, with no need to create survey weights as intermediate quantities.

- If you have conducted a survey and want to create weights for others to use: In this case it could make sense to anticipate the methods discussed in the present paper when forming the weights. It could help future users of the survey if the weights contain relevant information to help the model-based analysis perform well. Some research is needed here, given that probabilities of inclusion in the sample are generally not known, only estimated, and also given that the goal is adjustment to the population, not estimation of inclusion probabilities.

- If you are analyzing a survey collected by others where the weights have been supplied: Here, we hope our theoretical and applied examples give some sense of when it would make sense to follow the approach presented here.

References

Ansolabehere, S., Schaffner, B., and Luks, S. (2019). Guide to the 2018 Cooperative Congressional Election Survey. `https://cces.gov.harvard.edu/`

Bang, H., and Robins, J. M. (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics* **61**, 962–973.

Beaumont, J. F. (2008). A new approach to weighting and inference in sample surveys. *Biometrika* **95**, 539–553.

Ben-Michael, E., Feller, A., and Hartman, E. (2024). Multilevel calibration weighting for survey data. *Political Analysis* **32**, 65–83.

Bertail, P., and Combris, P. (1997). Bootstrap généralisé d'un sondage. *Annales d'Économie et de Statistique* **46**, 49–83.

Bisbee, J. (2019). BARP: Improving Mister P using Bayesian additive regression trees. *American Political Science Review* **113**, 1060–1065.

Bollen, K. A., Biemer, P. P., Karr, A. F., Tueller, S., and Berzofsky, M. E. (2016). Are survey weights needed? A review of diagnostic tests in regression analysis. *Annual Review of Statistics and Its Application* **3**, 375–392.

Brick, J. M. (2013). Unit nonresponse and weighting adjustments: A critical review. *Journal of Official Statistics* **29**, 329–353.

Brick, J. M., and Tourangeau, R. (2017). Responsive survey designs for reducing nonresponse bias. *Journal of Official Statistics* **33**, 735–752.

Broniecki, P., Leemann, L., and Wüest, R. (2021). Improved multilevel regression with post-stratification through machine learning (autoMrP). *Journal of Politics* **84**, 597–601.

Chen, Ch., Duan, N., Meng, X. L., and Alegria, M. (2006). Power-shrinkage and trimming: Two ways to mitigate excess weights. *Proceedings of the Section on Survey Research Methods, American Statistical Association*, 2839–2846.

Chen, Ci., Wakefield, J., and Lumley, T. (2014). The use of sampling weights in Bayesian hierarchical models for small-area estimation. *Spatial and Spatio-temporal Epidemiology* **11**, 33–43.

Chen, Q., Elliott, M. R., Haziza, D., Sadju, Y., Ghosh, M., Little, R. J. A., Sedransk, J., and Thompson, M. (2017). Approaches to improving survey-weighted estimates. *Statistical Science* **32**, 227–248.

Chen, Q., Elliott, M. R., and Little, R. J. A. (2012). Bayesian inference for finite population quantiles from unequal probability samples. *Survey Methodology* **38**, 203–214.

Chen, Y., Li, P., and Wu, C. (2020). Doubly robust inference with nonprobability survey samples. *Journal of the American Statistical Association* **115**, 2011–2021.

Cohen, M. P. (1997). The Bayesian bootstrap and multiple imputation for unequal probability sample designs. Proceedings of the Survey Research Methods Section, American Statistical Association, 635–638.

Dever, J. A., and Valliant, R. (2010). A comparison of variance estimators for poststratification to estimated control totals. *Survey Methodology* **36**, 45–56.

Dong, Q., Elliott, M. R., and Raghunathan, T. E. (2014). A nonparametric method to generate synthetic populations to adjust for complex sampling design features. *Survey Methodology* **40**, 29–46.

DuMouchel, W. H., and Duncan, G. J. (1983). Using sample survey weights in multiple regression analysis of stratified samples. *Journal of the American Statistical Association* **78**, 535–543.

Elliott, M. R., and Little, R. J. A. (2000). Model-based alternatives to trimming survey weights. *Journal of Official Statistics* **16**, 191–209.

Elliott, M. R., and Valliant, R. (2017). Inference for nonprobability samples. *Statistical Science* **32**, 249–264.

Fay, R. E., and Herriot, R. A. (1979). Estimates of income for small places: An application of James-Stein procedures to census data. *Journal of the American Statistical Association* **74**, 269–277.

Gelman, A. (2007). Struggles with survey weighting and regression modeling (with discussion). *Statistical Science* **22**, 153–188.

Gelman, A. (2008). Scaling regression inputs by dividing by two standard deviations. *Statistics in Medicine* **27**, 2865–2873.

Gelman, A. (2018). Regularized prediction and poststratification. Statistical Modeling, Causal Inference, and Social Science, 19 May. `https://statmodeling.stat.columbia.edu/2018/05/19/regularized-prediction-poststratification-generalization-mister-p/`

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D., Vehtari, A., and Rubin, D. B. (2013). *Bayesian Data Analysis*, third edition. CRC Press.

Gelman, A., and Little, T. C. (1997). Poststratification into many categories using hierarchical logistic regression. *Survey Methodology* **23**, 127–135.

Gelman, A., and Little, T. C. (1998). Improving upon probability weighting for household size. *Public Opinion Quarterly* **62**, 398–404.

Gelman, A., Vehtari, A., Simpson, D., Margossian, C. C., Carpenter, B., Yao, Y., Kennedy, L., Gabry, J., Bürkner, P., and Modrák, M. (2020). Bayesian workflow. `https://arxiv.org/abs/2011.01808`

Ghitza, Y., and Gelman, A. (2013). Deep interactions with MRP: Election turnout and voting patterns among small electoral subgroups. *American Journal of Political Science* **57**, 762–776.

Ghitza, Y., and Gelman, A. (2020) Voter registration databases and MRP: Toward the use of large scale databases in public opinion research. *Political Analysis* **28**, 507–531.

Gopelrud, M. (2024). Re-evaluating machine learning for MRP given the comparable performance of (deep) hierarchical models. *American Political Science Review* **118**, 529–536.

Graubard, B. I., and Korn E. L. (2002). Inference for superpopulation parameters using sample surveys. *Statistical Science* **17**, 73–96.

Haziza, D., and Beaumont, J. F. (2017). Construction of weights in surveys: A review. *Statistical Science* **32**, 206–226.

Heeringa, S. G., West, B. T., and Berglund, P. A. (2017). *Applied Survey Data Analysis*, second edition. CRC Press.

Holt, D., and Smith, T. M. F. (1979). Post stratification. *Journal of the Royal Statistical Society A* **142**, 33–46.

Kalton, G. (1983). Models in the practice of survey sampling. *International Statistical Review* **51**, 175–188.

Kennedy, L. A., and Gelman, A. (2020). Year 15 Fragile Families survey weight adjustment. Princeton Center for Research on Child Wellbeing. `https://fragilefamilies.princeton.edu/sites/g/files/toruqf2001/files/ff_const_wgtsy15.pdf`

Kennedy, L. A., and Gelman, A. (2021). Know your population and know your model: Using model-based regression and poststratification to generalize findings beyond the observed sample. *Psychological Methods* **26**, 547–558.

Kish, L., (1992). Weighting for unequal P_i. *Journal of Official Statistics* **8**, 183–200.

Kolenikov, S., West, B. T., and Lugtig, P. J. (2020). A checklist for assessing the analysis documentation for public-use complex sample survey data sets. *Survey Statistician* **81**, 50–62.

Korn, E. L., and Graubard, B. I. (1999). *Analysis of Health Surveys*. New York: Wiley.

Kuriwaki, S., Ansolabehere, S., Dagonel, A., and Yamauchi, S. (2024). The geography of racially polarized voting: Calibrating surveys at the district level. *American Political Science Review*

118, 922–939.

Lahiri, P., and Mukherjee, K. (2007). On the design-consistency property of hierarchical Bayes estimators in finite population sampling. *Annals of Statistics* **35**, 724–737.

Lei, R., Gelman, A., and Ghitza, Y. (2017). The 2008 election: A preregistered replication analysis. *Statistics and Public Policy* **4** (1), 1–8.

Léon-Novelo, L. G., and Savitsky, T. D. (2019). Fully Bayesian estimation under informative sampling. *Electronic Journal of Statistics* **13**, 1609–1645.

Li, K., and Si, Y. (2024). Embedded multilevel regression and poststratification: Model-based inference with incomplete auxiliary information. *Statistics in Medicine* **43**, 256–278.

Little, R. J. A. (1991). Inference with survey weights. *Journal of Official Statistics* **7**, 405–424.

Little, R. J. A. (1993). Post-stratification: A modeler's perspective. *Journal of the American Statistical Association* **88**, 1001–1012.

Little, R. J. A. (2004). To model or not to model? Competing modes of inference for finite population sampling. *Journal of the American Statistical Association* **99**, 546–556.

Little, R. J. A. (2015). Calibrated Bayes, an inferential paradigm for official statistics in the era of big data. *Statistical Journal of the International Association for Official Statistics* **31**, 555–563.

Little, R. J. A., and Vartivarian, S., (2003). On weighting the rates in non-response weights. *Statistics in Medicine* **22**, 1589–1599.

Little, T. C., and Gelman, A. (1998). Modeling differential nonresponse in sample surveys. *Sankhya* **60**, 101–126.

Lohr, S. (2022). *Sampling: Design and Analysis*, third edition. London: CRC Press.

Lopez-Martin, J., Phillips, J. H., and Gelman, A. (2022). Multilevel regression and poststratification case studies. `https://bookdown.org/jl5522/MRP-case-studies/`

Lumley, T. (2004). Analysis of complex survey samples. *Journal of Statistical Software* **9** (8), 1–19.

Lumley, T., and Scott, A. (2017). Fitting regression models to survey data. *Statistical Science* **32**, 265–278.

Maiti, T., Ren, H., and Sinha, S. (2014). Prediction error of small area predictors shrinking both means and variances. *Scandinavian Journal of Statistics* **41**, 775–790.

Makela, S., Si, Y., and Gelman, A. (2018). Bayesian inference under cluster sampling with probability proportional to size. *Statistics in Medicine* **37**, 3849–3868.

Meng, X. L. (1994). Multiple-imputation inferences with uncongenial sources of input. *Statistical Science* **9**, 538–558.

Miratrix, L. W., Sekhon, J. S., and Yu, B. (2013). Adjusting treatment effect estimates by post-stratification in randomized experiments. *Journal of the Royal Statistical Society B* **75**, 369–396.

Morikawa, K., Terada, Y., and Kim, J. K. (2022). Semiparametric adaptive estimation under informative sampling. `https://arxiv.org/abs/2208.06039/`.

O'Muircheartaigh, C., and Hedges, L. V. (2014). Generalizing from unrepresentative experiments: A stratified propensity score approach. *Journal of the Royal Statistical Society C* **63**, 195–210.

Pfeffermann, D. (1993). The role of sampling weights when modeling survey data. *International Statistical Review* **61**, 317–337.

Pfeffermann, D. (2011). Modelling of complex survey data: Why model? Why is it a problem? How can we approach it? *Survey Methodology* **37**, 115–136.

Pfeffermann, D., Skinner, C., Holmes, D., Goldstein, H., and Rasbash, J. (1998). Weighting for

unequal selection probabilities in multilevel models. *Journal of the Royal Statistical Society B* **60**, 23–40.

Potthoff, R., Woodbury, M., and Manton, K. (1992). "Equivalent sample size" and "equivalent degrees of freedom" refinements for inference using survey weights under superpopulation models. *Journal of the American Statistical Association* **87**, 383–396.

Rabe-Hesketh, S., and Skrondal, A. (2006). Multilevel modelling of complex survey data. *Journal of the Royal Statistical Society Series A: Statistics in Society* **169**, 805–827.

Rao, J. N. K. (2003). *Small Area Estimation.* New York: Wiley.

Rosenman, E. T. R., McCartan, C., and Olivella, S. (2023). Recalibration of predicted probabilities Using the "logit shift": Why does it work, and when can it be expected to work well? *Political Analysis* **31**, 651–661.

Rubin, D. B. (1976). Inference and missing data. *Biometrika* **63**, 581–592.

Rubin, D. B. (1996). Multiple imputation after 18+ years (with discussion). *Journal of the American Statistical Association* **91**, 473–520.

Särndal, C. E. (1978). Design-based and model-based inference in survey sampling (with discussion). *Scandinavian Journal of Statistics* **5**, 27–52.

Savitsky, T. D., Gershunskaya, J., and Crankshaw, M. (2022). Joint point and variance estimation under a hierarchical Bayesian model for survey count data. `https://arxiv.org/abs/2210.14366/`.

Schenker, N. (1985). Qualms about bootstrap confidence intervals. *Journal of the American Statistical Association* **80**, 360–361.

Si, Y., Pillai, N., and Gelman, A. (2015). Bayesian nonparametric weighted sampling inference. *Bayesian Analysis* **10**, 605–625.

Si, Y., Trangucci, R., Gabry, J., and Gelman, A. (2020). Bayesian hierarchical weighting adjustment and survey inference. *Survey Methodology* **46**, 181–214.

Si, Y., and Zhou, P. (2021). Bayes-raking: Bayesian finite population inference with known margins. *Journal of Survey Statistics and Methodology* **9**(4), 833–855.

Skinner, C. J. (1994). Sample models and weights. *Proceedings of the Section on Survey Research Methods, American Statistical Association*, 133-–142.

Solon, G., Haider, S. J., and Wooldridge, J. M. (2015). What are we weighting for? *Journal of Human Resources* **50**, 301–316.

Stanek, E. J., and Singer, J. M. (2004). Predicting random effects from finite population clustered samples with response error. *Journal of the American Statistical Association* **99**, 1119–1130.

Su, Y. S., and Gelman, A. (2023). Who wants school vouchers in America? A comprehensive study using multilevel regression and poststratification. *Social Sciences* **12**, 430.

Sugasawa, S., Tamae, H., and Kubokawa, T. (2017). Bayesian estimators for small area models shrinking both means and variances. *Scandinavian Journal of Statistics* **44**, 150–167.

Vehtari, A., Simpson, D., Gelman, A., Yao, Y., and Gabry, J. (2015). Pareto smoothed importance sampling. `https://arxiv.org/abs/1507.02646/`.

Voss, D. S., Gelman, A., and King, G. (1995). Pre-election survey methodology: Details from nine polling organizations, 1988 and 1992. *Public Opinion Quarterly* **59**, 98–132.

West, B. T., and McCabe, S. E. (2012). Incorporating complex sample design effects when only final survey weights are available. *Stata Journal* **12**, 718–725.

West, B. T., Sakshaug, J. W., and Aurelien, G. A. S. (2016). How big of a problem is analytic error in secondary analyses of survey data? *PLoS One* **11**, e0158120.

Winship, C., and Radbill, L. (1994). Sampling weights and regression analysis. *Sociological Methods and Research* **23**, 230–257.

Xie, H., Barker, L. E., and Rolka, D. B. (2020). Incorporating design weights and historical data into model-based small-area estimation. *Journal of Data Science* **18**, 115–131.

A. Examples with data and code

We give data and code for two examples: (1) Simulated data of a hypothetical marketing study, with a high correlation between the sampling weight and the outcome of interest; (2) Real data from an opinion poll (the Cooperative Congressional Election Study), incorporating sampling weights into a multilevel model used to estimate state-level attitudes to abortion from a national poll.

The examples are coded in R and Stan and demonstrate the integration of the steps of multilevel regression, poststratification, and adjustment for sampling weights. An area for future research and development is in cleaning the code, in particular making it easier for users to adapt their existing MRP code to implement the new MRPW approach.

A.1. Simulated-data example with code: logistic regression

We go through the hypothetical marketing study described in Section 5.1, simulating data, fitting the models, applying the poststratification procedure, and comparing the resulting inferences to the population values that were assumed in the simulation. The population comprises the N users who visit a webpage during a month, a small fraction of whom are given the opportunity to buy a certain product. Each user i has a customer engagement score, x_i, and the probability π_i of user i being exposed to the promotion is determined by an algorithm that depends on x_i and some other factors. The distribution of x_i is known in the population, and we know the values of π_i for the n users in the sample. Label y_i as the binary outcome of whether person i in the sample buys the product. The goal is to estimate the proportion of people in the population who would buy the product, and also to learn how this probability varies with x.

A.1.1. Simulating the population and sample

Here is code to simulate the data and sampling probabilities in the populations:

```
library("arm")
library("rstanarm")
library("cmdstanr")
set.seed(123)
N <- 1e6
x <- sample(1:10, size=N, replace=TRUE)
prob <- 1e-4 * x * exp(rnorm(N, 0, 0.8))
y <- ifelse(runif(N) < invlogit(0.9 + 0.1*x + 0.5*log(prob), 1, 0)
```

We calculate the quantity of interest, the true buy rate in the population:

```
> print(mean(y))
[1] 0.100
```

We then draw the sample:

```
in_sample <- (runif(N) < prob)
n <- sum(in_sample)
```

Here is the sample size:

```
> print(n)
[1] 771
```

We next prepare the data that would be available to the analyst:

```
w <- 1/prob                # inverse probability of selection
w <- w/mean(w[in_sample])  # normalized to have mean 1 in the data
sample_data <- data.frame(x, y, w)[in_sample,]
```

And here is the poststratification table:

```
poststrat <- data.frame(x = as.numeric(names(table(x))), N = as.numeric(table(x)))
J <- nrow(poststrat)
```

We then label the cells in the sample data:

```
sample_data$cell <- NA
for (j in 1:J) {
  sample_data$cell[sample_data$x==poststrat$x[j]] <- j
}
```

Now it is the analyst's turn. The starting point is to compute the raw and weighted estimates from the sample:

```
> print(c(mean(sample_data$y), mean(sample_data$w*sample_data$y)/mean(sample_data$w)))
[1] 0.163 0.103
```

As expected, the raw estimate is off—in this case, it is too high, which makes sense because in this simulation, the probability π of inclusion in the sample is positively correlated with the outcome, y. The weighted average is fine as a point estimate of the population average, but the analyst is also interested in how the outcome varies with x, hence the need for a regression model that adjusts for the sampling probabilities.

A.1.2. Estimating the model of weights in the population

Uncertainty in our inferences for the population will come from two sources: the fitted data model (2) and the fitted weight model (3). There is also potential uncertainty in the N_j's but we are ignoring any imperfections in the poststratification in this paper.

To show the basic idea, we start with a point estimate for the model of weights in the population. We return later in this section to account for uncertainty in that part of the model.

We start with a lognormal regression for the weights, as described in Section 3.1.

```
sample_data$v <- log(sample_data$w)
fit_v <- lm(v ~ x, data=sample_data)
display(fit_v)
```

Here is the result:

```
            coef.est coef.se
(Intercept)  0.81     0.09
x           -0.17     0.01
---
n = 771, k = 2
residual sd = 0.82, R-Squared = 0.20
```

The residual standard deviation σ is estimated at 0.82, so, following the mathematics in Section 3.1, the estimated distribution of v in the population is shifted to the right by σ^2, yielding the estimated distribution, $v|x \sim \text{normal}(0.81 - 0.17x + 0.82^2,\ 0.82)$.

A.1.3. Estimating the model of $y|x, v$

We next use the sample data to estimate the regression of the outcome on the predictor x and the log weights, v. In this case the outcome is binary and we fit logistic regression. We perform Bayesian inference so that we will have posterior simulation draws that capture inferential uncertainty.

```
fit_y <- stan_glm(y ~ x + v + x:v, family=binomial(link="logit"), data=sample_data)
print(fit_v, digits=2)
```

This yields:

```
            Median MAD_SD
(Intercept) -3.09   0.43
x            0.15   0.06
v           -0.39   0.44
x:v         -0.02   0.05
```

In order to use this model to make predictions, we need the distribution of v, given x, which we estimated in Section A.1.2.

A.1.4. Averaging over the estimated population distribution of $v|x$

For each poststratification cell j, we take 1000 draws from the fitted distribution of log weights, w, and then pipe these through the uncertainty in the fitted model for $y|x, w$ as represented by the S draws from that posterior distribution:

```
sims_y <- as.matrix(fit_y)
S <- nrow(sims_y)
L <- 1000
Ey <- array(NA, c(S, J))
for (j in 1:J){
  X_j <- poststrat$x[j]
  v <- rnorm(L, coef(fit_v) %*% c(1, X_j) + sigma(fit_v)^2, sigma(fit_v))
  Ey_pop <- posterior_epred(fit_y, newdata=data.frame(x=X_j, v))
  Ey[,j] <- rowMeans(Ey_pop)
}
```

The result is `Ey`, an $S \times J$ matrix, which contains S draws from the posterior distribution of $\mathrm{E}(y|X_j)$ for the J cells.

We can look at the inferences for the cells individually:

```
Ey_cell_est <- colMeans(Ey)
Ey_cell_se <- apply(Ey, 2, sd)
```

This gives estimates and posterior standard deviations for the J cells.

We can also compute the posterior distribution of the poststratified population average:

```
Ey_poststrat <- (Ey %*% poststrat$N) / sum(poststrat$N)
cat(mean(Ey_poststrat), "+/-", sd(Ey_poststrat), "\n")
```

which yields,

```
0.095 +/- -.012
```

A.1.5. Accounting for uncertainty in the fitted model for the weights

With a bit more effort, we can propagate the uncertainty in the estimated distribution of weights. This approach could be especially useful when weights are being estimated for non-probability samples (Chen et al., 2020). This requires computing posterior simulations for the parameters in the regression model for v:

```
fit_v <- stan_glm(v ~ x, data=sample_data, refresh=0)
sims_v <- as.matrix(fit_v)
```

and then looping the predictive calculations over the S simulation draws:

```
Ey <- array(NA, c(S, J))
for (s in 1:S){
  for (j in 1:J){
    X_j <- poststrat$x[j]
    v <- rnorm(L, sims_v[s,1:2] %*% c(1, X_j) + sims_v[s,"sigma"]^2, sims_v[s,"sigma"])
    Ey_pop <- invlogit(sims_y[s,1:4] %*% rbind(1, X_j, v, X_j*v))
    Ey[s,j] <- mean(Ey_pop)
  }
}
```

The way the model is set up, the parameters of the two regression models are independent in the posterior distribution, so we could use any ordering of simulation draws when propagating uncertainty, as long as we are consistent. For simplicity in setting up the code we use the same ordering of the S draws from the two fitted models.

We can then summarize the simulations in `Ey` as before. In this case, the results are the same as before to two decimal places (0.095 ± 0.012), which implies that the uncertainty for these population summaries is dominated by the uncertainty in the fitted regression of y.

A.1.6. Fitting a regression of log weights with normal-mixture error term

In this case, the log weights were simulated from a regression model with normal errors, but in general we would not know this. Following Section 3.2, we fit a linear regression for $v|x$ with an error term that is a mixture of three normals. We do this in Stan, and here is the program, which we call `mixture.stan`:

```
data {
  int M;
  int N;
  int K;
  vector[N] v;
  matrix[N,K] X;
}
parameters {
  vector[K] beta;
  simplex[M] lambda;
  ordered[M] mu;
  vector<lower=0>[M] sigma;
  real<lower=0> log_sigma_0;
}
model {
  vector[N] Xbeta = X*beta;
  lambda ~ lognormal(log(1./M), 1);
  mu ~ normal(0, 10);
  sigma ~ lognormal(log_sigma_0, 1);
  sum(lambda.*mu) ~ normal(0, 0.01);
  for (n in 1:N){
    vector[M] lps = log(lambda);
    for (m in 1:M){
      lps[m] += normal_lpdf(v[n] | Xbeta[n] + mu[m], sigma[m]);
    }
    target += log_sum_exp(lps);
  }
}
```

```
generated quantities {
  real mu_total = sum(lambda.*mu);
  real sigma_total = sqrt(sum(lambda.*((mu - mu_total)^2 + sigma^2)));
}
```

The model includes weakly informative priors on the parameters of the mixture components, and the line `sum(lambda.*mu) ~ normal(0, 0.01);` serves as a soft constraint to pin the mean of the fitted mixture model to zero, which allows the intercept of the regression to have the same interpretation as before.

We then fit the model in R and print and extract the results:

```
mixture <- cmdstan_model("mixture.stan")
M <- 3
K <- 2
mixture_data <- list(v=sample_data$v, X=cbind(rep(1,n), sample_data$x), N=n, K=K, M=M)
fit_v_mixture <- mixture$sample(data=mixture_data, seed=123, chains=4, parallel_chains=4)
print(fit_v_mixture, max_rows=20)
```

Here is the output:

```
   variable    mean  median   sd  mad      q5     q95 rhat ess_bulk ess_tail
 lp__        -948.41 -948.02 2.81 2.71 -953.58 -944.57 1.00      990     1338
 beta[1]        0.81    0.81 0.09 0.09    0.66    0.97 1.00     2137     1624
 beta[2]       -0.17   -0.17 0.01 0.01   -0.19   -0.15 1.00     2162     1697
 lambda[1]      0.30    0.23 0.24 0.22    0.04    0.78 1.01      889     1023
 lambda[2]      0.38    0.34 0.25 0.29    0.05    0.82 1.00     1177     1075
 lambda[3]      0.31    0.24 0.24 0.23    0.04    0.78 1.01     1166     1585
 mu[1]         -0.46   -0.43 0.27 0.31   -0.95   -0.09 1.00      887     1466
 mu[2]         -0.03   -0.02 0.26 0.19   -0.49    0.42 1.01     1034     1399
 mu[3]          0.50    0.47 0.31 0.34    0.09    1.04 1.00     1341     1636
 sigma[1]       0.66    0.69 0.18 0.14    0.32    0.90 1.00      931      683
 sigma[2]       0.73    0.77 0.18 0.12    0.38    0.97 1.00      907      635
 sigma[3]       0.65    0.67 0.15 0.14    0.36    0.85 1.00     1607     1809
 log_sigma_0   -0.43   -0.43 0.61 0.63   -1.46    0.56 1.00     2672     2552
 mu_total       0.00    0.00 0.01 0.01   -0.02    0.02 1.00     3888     2852
 sigma_total    0.82    0.82 0.02 0.02    0.79    0.86 1.00     4134     2662
```

A.1.7. Averaging over the fitted normal-mixture regression model for log weights

Next we extract the relevant parameters from the simulations and, for each simulation draw s and each poststratification cell j, we then take L draws v from the reweighted mixture model (8) and, for each, compute the expected value of y:

```
lambda_v <- as.matrix(fit_v_mixture$draws("lambda", format="df"))[,1:M]
mu_v <- as.matrix(fit_v_mixture$draws("mu", format="df"))[,1:M]
beta_v <- as.matrix(fit_v_mixture$draws("beta", format="df"))[,1:K]
sigma_v <- as.matrix(fit_v_mixture$draws("sigma", format="df"))[,1:M]
Ey <- array(NA, c(S, J))
for (s in 1:S){
  for (j in 1:J){
    X_j <- poststrat$x[j]
    v_hat_mixture <- as.numeric(beta_v[s,] %*% c(1, X_j)) + mu_v[s,]
    lambda_new <- lambda_v[s,] * exp(v_hat_mixture + 0.5*sigma_v[s,]^2)
    m <- sample(1:M, L, replace=TRUE, prob=lambda_new)
    v <- rnorm(L, v_hat_mixture[m] + sigma_v[s,m]^2, sigma_v[s,m])
```

```
    Ey_pop <- invlogit(sims_y[s,1:4] %*% rbind(1, X_j, v, X_j*v))
    Ey[s,j] <- mean(Ey_pop)
  }
}
```

As before, we can average over the cells to get S simulations of the poststratified population mean and compute its posterior mean and standard deviation:

```
Ey_poststrat <- (Ey %*% poststrat$N) / sum(poststrat$N)
cat(mean(Ey_poststrat), "+/-", sd(Ey_poststrat), "\n")
```

which yields,

```
0.095 +/- 0.012
```

This is approximately the same result as before, which makes sense given that the data were simulated from a model with a normal distribution for errors, which is approximately recovered by the fit of a mixture of three normals.

A.1.8. Bootstrapping residuals from the model for log weights

Finally we apply the simpler and perhaps more robust approach of Section 3.5 to simulate $v|x$ using a weighted bootstrap of the residuals:

```
fit_v <- lm(v ~ x, data=sample_data)
Ey <- array(NA, c(S, J))
for (j in 1:J){
  x_j <- poststrat$x[j]
  dist <- sample_data$x != x_j  # We have only one predictor so no need to sum over k
  same_cell <- dist==0
  c <- 1/sum(1/dist[!same_cell])
  weight_boot <- ifelse(same_cell, 1/30, c/dist) * exp(resid(fit_v))
  r_boot <- sample(resid(fit_v), L, replace=TRUE, prob=weight_boot)
  v <- predict(fit_v, newdata=data.frame(x=x_j)) + r_boot
  Ey_pop <- posterior_epred(fit_y, newdata=data.frame(x=x_j, v))
  Ey[,j] <- rowMeans(Ey_pop)
}
```

As in Section A.1.4, we use the point estimate of the fitted model of $v|x$—this is implied by the use of the `predict()` function for v—in order to keep the code cleaner, because we found that propagating uncertainty in that part of the model did not have any noticeable impact on the final results.

As before, we then summarize to obtain a posterior distribution for the population mean:

```
Ey_poststrat <- (Ey %*% poststrat$N) / sum(poststrat$N)
cat(mean(Ey_poststrat), "+/-", sd(Ey_poststrat), "\n")
```

which yields,

```
0.096 +/- 0.012
```

A.1.9. Integrated Bayesian computation

We can follow the plan described in Section 3.6 and embed all the computation inside a single Stan program, which we call `normal_logit_weighting_bootstrap.stan`. As indicated by the name of the file, this program fits a linear regression, $p(v|x, \phi)$, and a logistic regression, $p(y|x, v, \theta)$, and then in uses a weighted bootstrap to adjust for the unequal sampling probabilities in the generated quantities block:

```
 data {
  int N;  // Number of data points
  int K;  // Number of regression predictors
  int J;  // Number of poststratification cells
  array[N] int<lower=0,upper=1> y;    // Binary outcome
  array[N] int<lower=0,upper=J> cell; // Poststratification cells of data
  vector<lower=0>[N] w;    // Sampling weights (data with 0 or neg weights must be removed)
  matrix[N,K] X;           // Regression predictors (including constant term)
  matrix[J,K] X_poststrat; // Regression predictors for poststratification cells
  vector[J] N_poststrat;   // Sizes of poststratification cells
  int L;  // Number of simulations for approximating p(v|x)
}
transformed data {
  vector[N] v = log(w);
  matrix[N,2*K] Xv = append_col(X, X .* rep_matrix(v, K));
  vector[K] dist_sd;
  matrix[N,J] dist, a;
  vector[J] c;
  for (k in 1:K){
    dist_sd[k] = sd(col(X_poststrat,k));
  }
  for (j in 1:J){
    c[j] = 0;
    for (i in 1:N){
      dist[i,j] = 0;
      for (k in 1:K){
        dist[i,j] = dist[i,j] + X[i,k] == X_poststrat[j,k];
      }
      if (dist[i,j] > 0) c[j] = c[j] + 1/dist[i,j];
    }
    for (i in 1:N){
      if (dist[i,j] == 0) a[i,j] = 1./30.;
      else a[i,j] = c[j]/dist[i,j];
    }
  }
}
parameters {
  vector[K] b_v;    // Coefs for regression of v on X
  vector[2*K] b_y;  // Coefs for regression of y on X interacted with v
  real<lower=0> sigma_v;  // Residual sd of regression of v
}
transformed parameters {
  vector[N] E_v = X*b_v;
}
model {
  v ~ normal(E_v, sigma_v);
```

```
  y ~ bernoulli_logit(Xv*b_y);
}
generated quantities {
  vector[J] E_y_poststrat;
  real E_y_poststrat_mean;
  {
  vector[N] resid = v - E_v;
  vector[J] v_pred = X_poststrat * b_v;
  for (j in 1:J){
    vector[N] prob_boot = col(a,j) .* exp(resid);
    prob_boot = prob_boot/sum(prob_boot);
    vector[L] resid_boot;
    for (l in 1:L) {
      resid_boot[l] = resid[categorical_rng(prob_boot)];
    }
    vector[L] v_sim = v_pred[j] + resid_boot;
    matrix[L,K] X_sim = rep_matrix(X_poststrat[j], L);
    matrix[L,2*K] Xv_sim = append_col(X_sim, X_sim .* rep_matrix(v_sim, K));
    vector[L] E_y_sim = inv_logit(Xv_sim * b_y);
    E_y_poststrat[j] = mean(E_y_sim);
  }
  E_y_poststrat_mean = sum(N_poststrat .* E_y_poststrat) / sum(N_poststrat);
  }
}
```

We run the Stan program from R:

```
integrated_boot <- cmdstan_model("normal_logit_weighting_bootstrap.stan")
integrated_data <- list(N=n, K=2, J=J, L=100, y=sample_data$y, w=sample_data$w,
  X=cbind(rep(1,n),sample_data$x), X_poststrat=cbind(rep(1,J),poststrat$x),
  N_poststrat=poststrat$N, cell=sample_data$cell)
integrated_boot_fit <- integrated_boot$sample(integrated_data, seed=123, chains=4,
  parallel_chains=4)
print(integrated_boot_fit, c("b_v", "b_y", "sigma_v", "E_y_poststrat",
  "E_y_poststrat_mean"), max_rows=30, digits=3)
```

When coding directly in Stan, this runs much faster than our earlier indirect approach piping the simulations of $v|x$ through the posterior prediction functions in rstanarm or brms. Here is the output:

```
           variable   mean median    sd   mad     q5    q95  rhat ess_bulk ess_tail
 b_v[1]               0.811  0.811 0.091 0.089  0.660  0.965 1.002     2419     2494
 b_v[2]              -0.173 -0.173 0.012 0.012 -0.193 -0.153 1.003     2359     2253
 b_y[1]              -3.135 -3.114 0.446 0.425 -3.887 -2.427 1.001     1749     1682
 b_y[2]               0.152  0.150 0.060 0.060  0.056  0.251 1.002     1747     1506
 b_y[3]              -0.429 -0.413 0.444 0.436 -1.203  0.281 1.000     1913     1890
 b_y[4]              -0.015 -0.017 0.055 0.054 -0.102  0.080 1.001     1837     1796
 sigma_v              0.817  0.816 0.021 0.021  0.784  0.853 1.000     3288     2492
 E_y_poststrat[1]     0.037  0.031 0.023 0.019  0.011  0.080 1.000     1630     1746
 E_y_poststrat[2]     0.040  0.036 0.021 0.018  0.014  0.078 1.000     1657     1524
 E_y_poststrat[3]     0.048  0.046 0.018 0.017  0.023  0.080 1.000     1706     1707
 E_y_poststrat[4]     0.058  0.056 0.015 0.015  0.035  0.085 1.000     1889     2039
 E_y_poststrat[5]     0.074  0.073 0.014 0.014  0.053  0.098 1.001     2253     2353
 E_y_poststrat[6]     0.089  0.088 0.013 0.013  0.068  0.111 1.000     2808     2463
 E_y_poststrat[7]     0.113  0.112 0.014 0.013  0.091  0.136 1.001     3866     3284
```

```
E_y_poststrat[8]     0.128  0.127 0.016 0.016  0.102  0.155 1.000     4048     3451
E_y_poststrat[9]     0.170  0.170 0.020 0.021  0.137  0.203 1.001     3279     3362
E_y_poststrat[10]    0.200  0.200 0.028 0.028  0.157  0.247 1.000     2766     2957
E_y_poststrat_mean   0.096  0.095 0.012 0.012  0.077  0.117 1.000     3078     2857
```

This gives us inference for both sets of regression parameters, the residual standard deviation of the regression of $v|x$, population averages within all the poststratification cells, and the same 0.096 ± 0.012 estimate for the poststratified population mean.

A.1.10. Estimating a population regression

Following the method explained in Section 2.2, we can use the posterior simulations of the poststratification cell averages to perform inference for the population regression of y on x. Here is the R code:

```
pop_coef <- array(NA, c(S, 2))   # 2 predictors (including the intercept)
cell_means <- integrated_boot_fit$draws(format="df", variables="E_y_poststrat")
for (s in 1:S){
  y_j <- as.numeric(poststrat$N * cell_means[s,])[1:J]
  fit <- glm(cbind(y_j, poststrat$N - y_j) ~ poststrat$x,
    family=quasibinomial(link="logit"))
  pop_coef[s,] <- coef(fit)
}
pop_coef_hat <- apply(pop_coef, 2, mean)
pop_coef_se <- apply(pop_coef, 2, sd)
```

The estimated intercept and slope are -3.67 and 0.22 with standard errors 0.58 and 0.07, respectively.

A.2. Real-data example with code: multilevel linear regression

We next show a small real-data example from our MRP case study (Lopez-Martin et al., 2022), which uses data from the 2018 Cooperative Congressional Election Study (Ansolabehere et al., 2019), a survey that included weights which for simplicity we had not included in our earlier case study. Here, we estimate opinion on abortion (using a composite response on a 0–6 scale which we treat as a continuous outcome), poststratifying on ethnicity, age, education, and state. The biggest difference between this and the previous example is that our goal here is inference for small areas—in this case, state-level estimates of abortion attitudes—rather than for the population average. We work with a random sample of 500 respondents so as to make the benefits of multilevel modeling more dramatic.

A.2.1. Simple unweighted and weighted MRP

First we set up R, read in the data, extract a subset, and renormalize the weights:

```
library("rstanarm")
source("setup_1.R")
set.seed(123)
n <- 500
subset <- sort(sample(nrow(df), n))
data <- df[subset,]
data$w <- data$w/mean(data$w)
```

We then fit the multilevel model with and without weights:

```
fit_unweighted <- stan_glmer(abortion ~ (1 | state) + (1 | eth) + (1 | educ) +
  (1 | age) + male + (1 | male:eth) + (1 | educ:age) + (1 | educ:eth) + repvote +
  (1 | region), data=data, cores=4)
```

```
fit_weighted <- stan_glmer(abortion ~ (1 | state) + (1 | eth) + (1 | educ) +
  (1 | age) + male + (1 | male:eth) + (1 | educ:age) + (1 | educ:eth) + repvote +
  (1 | region),  weight=w, data=data, cores=4)
```

In addition to the poststratification variables, the model includes two state-level predictors: an indicator for region and the Republican vote share in the state in the 2016 presidential election, standardized to have zero mean and unit standard deviation among the 50 states.

We look at the fitted models to make sure they make sense:

```
> print(fit_unweighted)
            Median MAD_SD
(Intercept)  2.6    0.5
male        -0.1    0.4
repvote      0.4    0.1

Auxiliary parameter(s):
      Median MAD_SD
sigma 1.9    0.1

Error terms:
 Groups   Name        Std.Dev.
 state    (Intercept) 0.22
 educ:age (Intercept) 0.53
 educ:eth (Intercept) 0.46
 male:eth (Intercept) 0.64
 age      (Intercept) 0.36
 educ     (Intercept) 0.39
 region   (Intercept) 0.47
 eth      (Intercept) 0.66
 Residual             1.90

> print(fit_weighted)
stan_glmer
 family:       gaussian [identity]
 formula:      abortion ~ (1 | state) + (1 | eth) + (1 | educ) +
    (1 | age) + male + (1 | male:eth) + (1 | educ:age) + (1 | educ:eth) +
    repvote + (1 | region)
  observations: 500
------
            Median MAD_SD
(Intercept)  2.6    0.5
male        -0.1    0.4
repvote      0.1    0.2

Auxiliary parameter(s):
      Median MAD_SD
sigma 2.6    0.1

Error terms:
 Groups   Name        Std.Dev.
 state    (Intercept) 0.19
 educ:age (Intercept) 0.32
 educ:eth (Intercept) 0.42
 male:eth (Intercept) 0.50
```

```
 age      (Intercept) 0.51
 educ     (Intercept) 0.42
 region   (Intercept) 0.72
 eth      (Intercept) 0.56
 Residual             2.62
```

Next we poststratify each fitted model to obtain inferences for the 50 states, which we graph vs. Republican vote share in the previous election:

```
poststrat_state_graph <- function(Ey, poststrat_df, statelevel_predictors_df, y_range,
    ylab="", main="") {
  S <- nrow(Ey)
  states <- names(table(poststrat_df$state))
  n_state <- length(states)
  mrp_state <- array(NA, c(S, n_state))
  for (i in 1:n_state){
    keep <- poststrat_df$state==states[i]
    mrp_state[,i] <- Ey[,keep] %*% poststrat_df$N[keep] / sum(poststrat_df$N[keep])
  }
  mrp_state_est <- apply(mrp_state, 2, mean)
  mrp_state_se <- apply(mrp_state, 2, sd)
  plot(range(statelevel_predictors_df$repvote), y_range,
    xlab="Standardized Republican vote share", ylab=ylab, bty="l", type="n", main=main)
  for (i in 1:n_state){
    lines(rep(statelevel_predictors_df$repvote[i], 2),
      mrp_state_est[i] + c(-1,1)*mrp_state_se[i], col="darkgray", lwd=.5)
  }
  text(statelevel_predictors_df$repvote, mrp_state_est, states, cex=.8, col="blue")
}
poststrat_state_graph(posterior_epred(fit_unweighted, newdata=poststrat_df),
  poststrat_df, statelevel_predictors_df, c(1.6,3.9),
  ylab="Avg abortion attitude (on 0-6 scale)", main="Unweighted MRP of y|x")
poststrat_state_graph(posterior_epred(fit_weighted, newdata=poststrat_df),
  poststrat_df, statelevel_predictors_df, c(1.6,3.9),
  ylab="", main="Weighted MRP of y|x")
```

The left and center plots on Figure 2 show the results. In this case, the difference between unweighted and weighted MRP inferences appear to come from estimation of the coefficients for region in the fitted models.

A.2.2. Estimating the model of weights in the population

We checked and this survey has no data with zero or negative weights, so we proceeded by fitting a regression of log weights on the predictors used in the MRP model:

```
data$v <- log(data$w)
fit_v <- stan_glmer(v ~ (1 | state) + (1 | eth) + (1 | educ) +
  (1 | age) + male + (1 | male:eth) + (1 | educ:age) + (1 | educ:eth) + repvote +
  (1 | region), data=data, cores=4)
print(fit_v)
```

Several of the demographic and geographic variables turn out to be strong predictors of the log weights, but the residuals remain far from zero (an estimated residual standard deviation of 0.6, with most of the residuals falling between -1 and 1; see the discussion of the example Section 6 for details.

A.2.3. Estimating the model of $y|x, v$

We begin the joint modeling by fitting a multilevel regression of the outcome given predictors, interacting everything with the log-weight variable, v:

```
fit_y <- stan_glmer(abortion ~ (1 + v | state) + (1 + v | eth) + (1 + v | educ) +
  (1 + v | age) + v*male + (1 + v | male:eth) + (1 + v | educ:age) +
  (1 + v | educ:eth) + v*repvote + (1 + v | region), data=data, cores=4)
print(fit_y)
```

which yields,

```
            Median MAD_SD
(Intercept)  2.6    0.3
v            0.1    0.3
male        -0.2    0.3
repvote      0.4    0.1
v:male      -0.3    0.4
v:repvote   -0.1    0.2

Auxiliary parameter(s):
      Median MAD_SD
sigma 1.9    0.1

Error terms:
 Groups   Name        Std.Dev. Corr
 state    (Intercept) 0.20
          v           0.32     -0.26
 educ:age (Intercept) 0.48
          v           0.29     -0.08
 educ:eth (Intercept) 0.39
          v           0.24     0.15
 male:eth (Intercept) 0.43
          v           0.35     0.15
 age      (Intercept) 0.22
          v           0.19     0.21
 educ     (Intercept) 0.22
          v           0.23     -0.01
 region   (Intercept) 0.30
          v           0.28     0.19
 eth      (Intercept) 0.40
          v           0.41     -0.02
 Residual             1.88
```

If there is interest, it would be possible to study the varying coefficients in this model to see which particular age categories, ethnic groups, ethnicities, regions, and states are associated with higher weights, which should comport with general understanding of which people are less likely to participate in surveys (Voss et al., 1995). Perhaps surprisingly, younger respondents do not have higher weights; perhaps the survey put in special effort to reach young people. The residual standard deviation of 1.89 is essentially unchanged from the 1.90 from the regression that did not include v (see the fitted model `fit_unweighted` on page 37), so in this case the weights are not highly predictive of the outcome. This is fine: our goal here is to demonstrate an approach that can be applied in general, not just to outcomes where the weighting makes a big difference.

A.2.4. Bootstrapping residuals from the model for log weights

We can simulate the distribution of v within each poststratification cell using a weighted bootstrap of residuals as described in Section 3.5. In this case, though, the model of $y|x, w$ is linear in v, hence all we need is $\mathrm{E}(v|x)$, so we can replace the bootstrap sample of residuals by the appropriate weighted average for each cell:

```
r <- resid(fit_v)
vars <- c("state", "eth", "educ", "age", "male")
K <- length(vars)
for (j in 1:J){
  dist <- rowSums(as.matrix(data[,vars]) !=
    matrix(rep(as.matrix(poststrat_df[j,vars]), n), ncol=K, byrow=TRUE))
  same_cell <- dist==0
  c <- 1/sum(1/dist[!same_cell])
  w_boot <- ifelse(same_cell, 1/30, c/dist) * exp(r)
  r_pop_mean[j] <- sum(w_boot*r)/sum(w_boot)
}
```

We then pipe this thorough our fitted model to obtain posterior simulation draws of the expected outcomes within all J cells:

```
v_pop_mean <- colMeans(posterior_epred(fit_v, newdata=poststrat_df)) + r_pop_mean
Ey <- posterior_epred(fit_y, newdata=data.frame(poststrat_df, v=v_pop_mean))
```

And then we plot the inferences for the 50 states. The result is the rightmost plot of Figure 2.

A.3. Real-data example with code: multilevel logistic regression

Section A.2 demonstrated our procedure in the context of MRP with a linear model fit to data from the 2018 Cooperative Election Study. We now illustrate a logistic model using a binary outcome from the same survey, in this case an indicator for whether the 0–6 abortion attitude response is at least 3. The procedure is mostly unchanged, but at the end there is a challenge: $\mathrm{E}(y|x, v)$ is now a nonlinear function of v, so we cannot just summarize the distribution of $v|x$ by its expectation. Instead, within each poststratification cell j, we create 1000 simulations $v^l|X_j$, then compute $\mathrm{E}(y|X_j, v^l)$ for each, then average over these to compute a Monte Carlo estimate of $\mathrm{E}(y|x, v_j)$. All this is conditional on the parameter vector θ^s; we average over the distribution induced by the S simulations to compute posterior mean and standerd deviations. As before, we condition on a point estimate of ϕ, the parameters governing the model for $v|x$, because uncertainty in ϕ seems to be a minor contributor to posterior uncertainty in $\mathrm{E}(y|x)$.

First we define a binary outcome and fit a multilevel logistic regression predicting it given x and v:

```
data$abortion3 <- data$abortion >= 3
fit_y_binary <- stan_glmer(abortion3 ~ (1 + v | state) + (1 + v | eth) +
  (1 + v | educ) + (1 + v | age) + v*male + (1 + v | male:eth) + (1 + v | educ:age) +
  (1 + v | educ:eth) + v*repvote + (1 + v | region),
  family=binomial(link="logit"), data=data, cores=4)
```

We then average over the fitted distribution of v within each poststratification cell. The number of poststratification cells is large enough, with each cell representing only a small fraction of the population, so we can simulate just $L = 1$ draw of v within each cell. The computation then becomes,

```
L <- 1  # The code is written for general L; here we use L=1.
Ey_alt <- array(NA, c(S, J))
Ev0 <- colMeans(posterior_epred(fit_v, newdata=poststrat_df))
```

```
poststrat_rep <- NULL
v_rep <- NULL
w_boot <- array(NA, c(J, n))
for (j in 1:J){
  dist <- rowSums(as.matrix(data[,vars]) !=
    matrix(rep(as.matrix(poststrat_df[j,vars]), n), ncol=K, byrow=TRUE))
  same_cell <- dist==0
  c <- 1/sum(1/dist[!same_cell])
  w_boot[j,] <- ifelse(same_cell, 1/30, c/dist) * exp(r)
}
for (l in 1:L){
  poststrat_rep <- rbind(poststrat_rep, poststrat_df)
  r_boot <- rep(NA, J)
  for (j in 1:J){
    r_boot[j] <- sample(r, 1, replace=TRUE, prob=w_boot[j,])
  }
  v_rep <- c(v_rep, Ev0 + r_boot)
}
```

We then compute S posterior draws of $\mathrm{E}(y|x)$ for the JL cells and average the L points within each poststratification cell to obtain an $S \times J$ matrix representing S posterior draws of average opinion within the J cells:

```
Ey_pop <- posterior_epred(fit_y_binary, newdata=poststrat_rep)
for (j in 1:J){
  indexes <- j + (0:(L-1))*J
  Ey_alt[,j] <- rowMeans(Ey_pop[,indexes, drop=FALSE])
}
```

From there, we can poststratify within states to get S posterior draws of average opinion within the 50 states. Figure 3 shows the result, compared to MRP and weighted MRP of the data.

As discussed in Section A.1.9, the best way to proceed would be to embed all the computation inside a single Stan program, extending `rstanarm` or `brms` to perform the necessary sampling of v, inference for $\mathrm{E}(y|x, v, \theta)$, and poststratification in a generated quantities block, at which point the results could be plotted as above.